1

Introduction

Network Analysis Today

1.1 WHY STRUCTURE MATTERS

Stop. Take a moment to look around. What do you see? No matter where you are, you are likely perceiving a world consisting of *things*. Maybe you are reading this book in a coffee shop, and if so, you probably see people, cups, books, chairs, and so on. You see a world of objects with properties, yourself included: white cups are on wooden tables, people sitting in chairs are reading books and talking with one another. At the same time, you are a subject, responding to this world and actively bringing yourself and these objects into interrelation. And yet, the world of objects with properties that you are perceiving is but one slice of a complex reality.

What is less obvious and often taken for granted is all the *relationships* that come together to make this world of things a sensible, navigable reality. In the coffee shop you are unlikely to notice the complex patterns of exchanges in resources that brought the coffee to your table, the hierarchy of relationships that organizes the work roles in the coffee shop, or the stable pattern of interactions among customers coming and going that make the coffee shop a hub in the flows in so many people's everyday lives. You take those exchanges and relationships for granted; and yet, you are embedded within them. You and the world of things you perceive are inextricably tied together through these invisible webs of flows, exchanges, and more or less stable relationships. They uphold and provide meaning for your subjectively experienced reality.

The natural and social worlds are filled with flows, exchanges, and relationships like these. By studying these largely unseen patterns, or *networks*, we come to understand myriad social phenomena – for example, how persons assume distinct roles, like barista, and the role relations between employees and patrons; the ways in which personal relationships form and evolve from that of employee–patron or coworkers to something more intimate, like friends or

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romantic partners; and how gossip spreads information across some of these role relations more than others. Many of our dearest social institutions are replete with persistent associations, such as peer groups, families, and schools. Even our casual dinner conversations can be viewed as having recognizable patterns that we interpret as either a positive bonding moment or an awkward one. These all entail *social* networks – that is, flows, exchanges, and relationships that exist only within human experience and behavior.

But take a moment to consider all the other phenomena that also have relational properties. Molecules are structures formed by an assortment of atomic bonds. Brains function through structures of neural connections. Ecosystems entail structures of food webs where various animals and plants consume one another, thereby creating flows of carbon and energy. The Internet is organized by links that connect web pages. Markets move in response to a system of patterned transactions. Language creates meaning by assembling a complex set of relationships between words, sentences, and grammar. These are also networks.

We cannot understand either the social or wider world without understanding relationships and the networks they form. These structures define the environments in which core scientific phenomena take place. They are not just background connections in the understanding of life, but are integral to explaining and modeling complex phenomena in accurate and meaningful ways. It is hard to imagine a discussion of brain functioning without references to brain regions and neural activity linking neurons and those regions. Likewise, it is hard to imagine studying the social aspects of life without examining actions and relationships that connect people. In short, the interconnectedness of objects is a fundamental property of the world, and makes the world possible. Regardless of the social actors or objects being connected (i.e., networked), the properties and dynamics of being interconnected are something all phenomena share. And this is what network analysis seeks to understand. Many disciplines and fields concern networks, and the specific *content* of these disciplines vastly differs, but it is the focus on *structure*s and the interdependencies to which structures give rise that unites them.

1.2 WHAT IS STRUCTURE?

> If we ever get to the point of charting a whole city or a whole nation, we would have ... a picture of a vast solar system of intangible structures, powerfully influencing conduct, as gravitation does in space. Such an invisible structure underlies society and has its influence in determining the conduct of society as a whole.
>
> J. L. Moreno, *New York Times*, April 13, 1933

In the most general terms, a structure is an arrangement of related objects that form a *pattern*. Patterns arise everywhere, but most remain largely unseen, discernable only at a physical or conceptual distance. This is especially true of *social structures*, which are patterns of interactions among people, such as

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those envisioned in the preceding quote by Jacob Moreno, a founder of the social network approach, or our envisioning of the coffee shop at the beginning of this chapter. In trying to discern social structures, people are a bit like fish in a school: each individual perhaps sees some fleeting aspects of structure, but always from a partial, subjective viewpoint. A more objective structural understanding requires the aid of tools that allow us to see beyond our own senses and cognitive limitations. As network researchers, we are in the business of devising such tools for understanding the world. Unlike fish, people have created schools of thought (forgive the pun) dedicated to the discovery, preservation, and transmission of knowledge and tools for addressing these problems.

The network tradition is one such school of thought. It stands in contrast to more traditional schools in the social and physical sciences that use tools focusing on individual objects and their characteristics. Such individual-level approaches have dominated entire disciplines in the social sciences, such as psychology and economics. Even in the clearly less individual-centered discipline of sociology, the primary tool for understanding behavior for decades was the survey. In seeking generalizability, surveys draw random samples of *individuals* from populations, thereby sacrificing most of the local structures of family, friends, coworkers, neighborhoods, and communities (McPherson & Smith 2019). In random survey designs, the connections among respondents violate statistical assumptions of independent observations and must therefore be avoided. An independent, random sample of persons is easy to collect and analyze, but this practice comes at a cost: by divorcing the respondents from their social context, the concepts and tools of traditional survey research treat each person as an isolated entity and regard their characteristics as having reified meaning. Such data tend to yield variable-based explanations for social phenomena in which individual characteristics, like age and education, are treated as causal factors (Abbott 1988).

Network scholars see the interdependencies among actors (i.e., my behavior is shaped by my relationships with others) not as a complication to avoid but instead as the subject of inquiry. Within these networks – and by virtue of their links and position – individual objects derive their meaning. A person is defined by their unique position and trajectory across networks over time (Mead 1934). By virtue of the network pattern, we also identify larger social constructs, like groups and roles. A community's internal process is in great part reflected by the patterns of associations that define them. In effect, the network of relations is a dualistic means of discerning what it means to be *both* individuals and groups, and it regards their definition in a contextualized, situated light. The exact objects of interest (e.g., people, animals, or airports) can vary widely across substantive fields, as can the relations, or links, that connect them (e.g., friendship, kinship, advice, fighting, or grooming). The interconnectedness of things makes them interdependent and reactive to one another. In short, when observed over time, objects affecting one another through ties are *systems*.

In general, we can think of systems as falling under four main types, with very different types of objects and links (Figure 1.1).

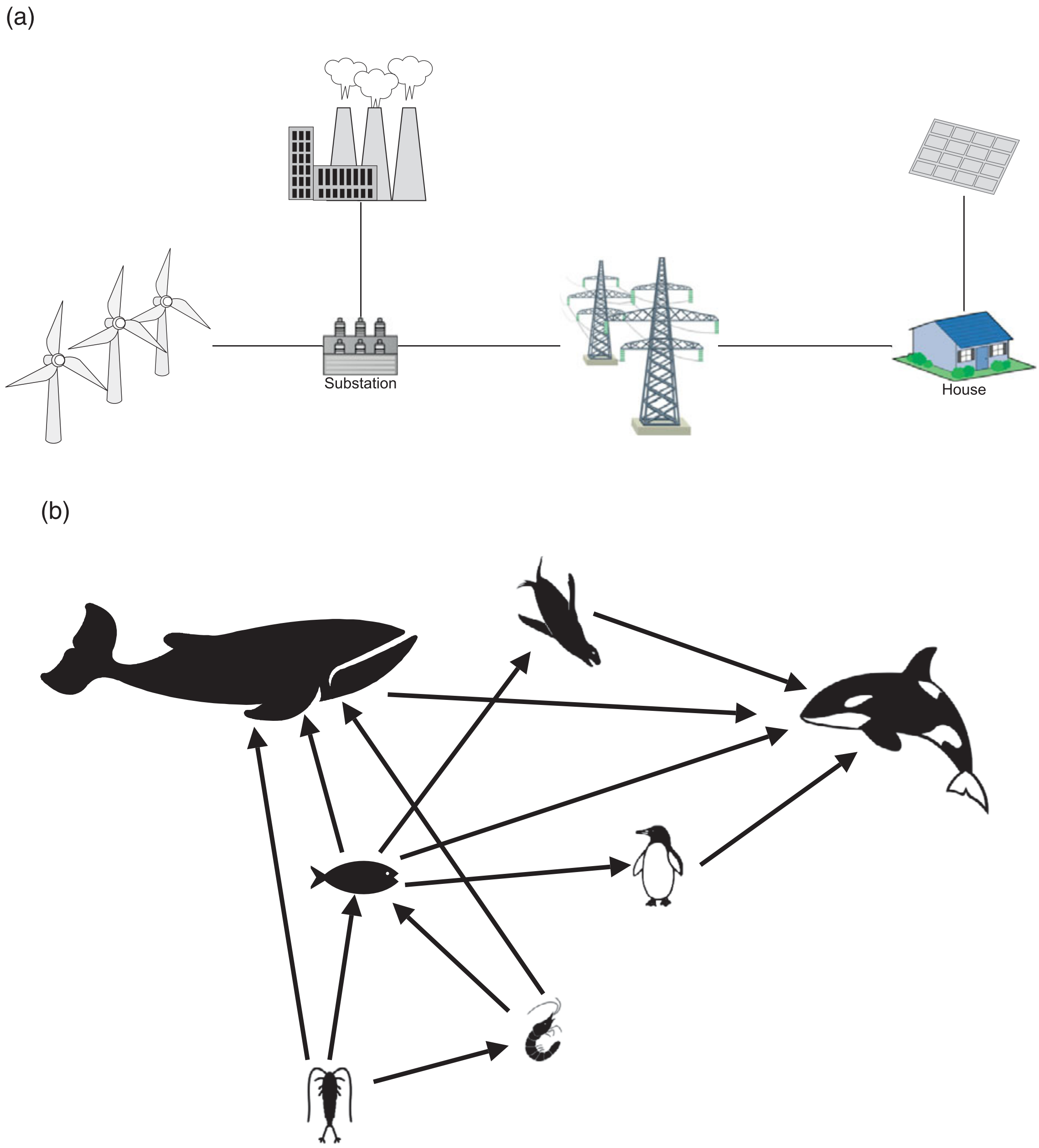

FIGURE 1.1 Types of systems. (a) *Mechanical systems* (machines): elements and their interactions are designed, tightly coupled, and restricted to efficiently achieve a goal. The electrical grid is a good example of a mechanical system. (b) *Living systems* (cells, bodies, ecosystems): elements are subsystems with a degree of autonomy, with communication and influence in multiple directions. These often include interactional feedback loops and adaptive learning processes. (c) *Social systems* (groups or larger collectivities): elements are often persons interrelated in patterns of exchange that reflect group memberships and hierarchies. These systems can vary in their differentiation and volatility. The Western European kinship system is a good example of a social system that organizes gender roles, such as being an aunt to a focal individual. (d) *Cultural systems* (interrelated meanings): elements are symbols that form semiotic systems through cognitive and affective connections of similarity and difference. Networks of semantic relations can be used to depict cultural systems, such as that organizing the classification of animals as mammals.

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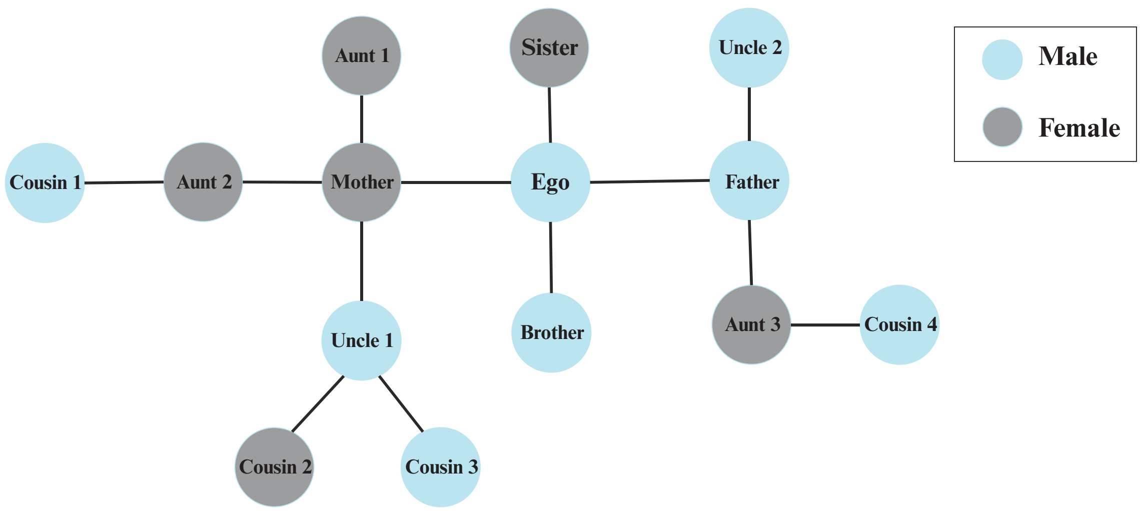

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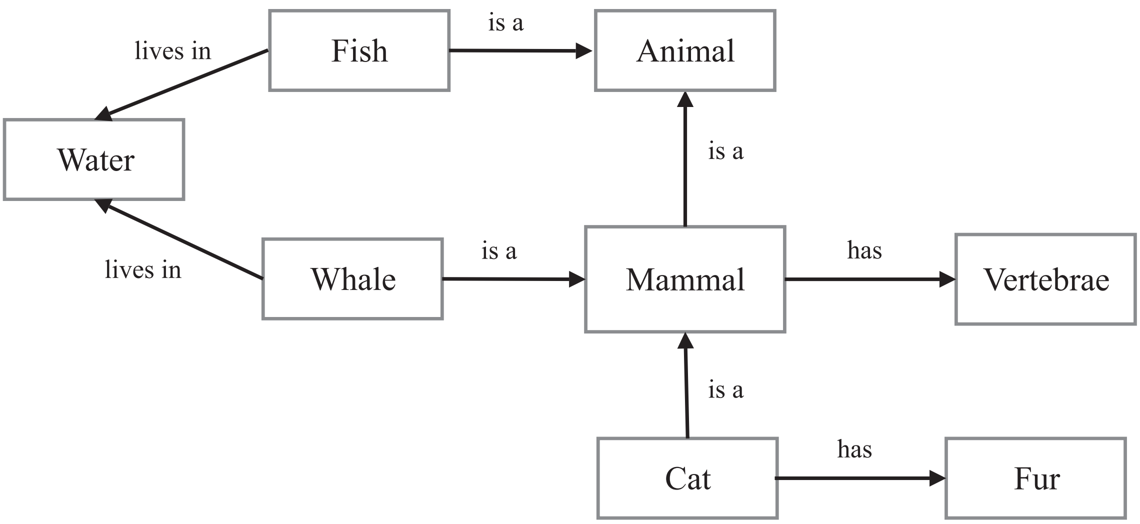

FIGURE 1.1 (*cont.*)

Tracing such systems can be conceptually and computationally challenging. Researchers from an array of fields have formulated conceptual and analytic tools to help us see structures and to understand their importance. In fact, since the 1940s, the field of *cybernetics* has been an interdisciplinary attempt to unite the sciences through the study of various systems. As with a variety of other scientific endeavors, network analysis seeks to better understand the underlying reality that our world is structured by overlapping and often complex systems.

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1.3 ORIGINS OF NETWORK ANALYSIS

Contributions to the origins of network analysis have come from a variety of other scientific domains with diverse analytical and theoretical orientations (see Freeman 2004 for a detailed history). Involved fields included graph theory (Euler 1736), sociology (Davis, Gardner, & Gardner 1941; Roethlisberger, Dickson, & Wright 1947 [1939]; Simmel 1909), education (Almack 1922), anthropology (Barnes 1954; Nadel 1957), and psychology (Heider 1946). In the early period, most of the effort was placed on developing a set of concepts, theories of tie formation (e.g., how individuals decide to become friends), and exploratory research on small groups ($N < 100$). It is in this era that fundamental theories and concepts emerged. Many of these early concepts and theories will be covered in this volume.

From the 1960s to the 1990s, the field witnessed a wide assortment of concurrent interdisciplinary work on social networks (Scott 2002), mostly in the disciplines of sociology, anthropology, and social psychology. Much of this work focused on larger samples of persons and groups ($N < 2{,}000$), such as clubs, schools, and organizations. This work used more complex methods to analyze social systems than previous work. What became known as The Harvard School is exemplary of this period and centered on the work and ideas of Harrison White, a scholar with PhDs in physics and sociology. Academics aligned with this school of thought created mathematical approaches to identifying structurally equivalent persons in graphs (Lorrain & White 1971), techniques for revealing network positions and their interrelations as role structures (White, Boorman, & Breiger 1976), and approaches to the study of affiliation networks – that is, ties that are based on belonging to the same groups or events (Breiger 1974). From this school emerged other scholars who established many of the core concepts used in network analysis today; examples include Mark Granovetter's (1973) notions of weak ties and structural embeddedness (1985), and Peter Blau's (1977) notion of structural differentiation. Much of this work extended the core ideas of the prior generation (e.g., Nadel 1957; Simmel 1909) by exploring their mathematical elaboration and operationalization using mathematical models and statistical tools. What resulted was a fruitful period in social network analysis that produced complex descriptive research on groups and their relations and introduced hypothesis testing.

With the advance of computing and the popularization of the Internet in the 1990s, the size and availability of network information exploded, and scholars from the fields of engineering and physics entered *en masse*. Social scientists now share the stage in the development of network analysis with computer scientists (e.g., Kleinberg 2000; Leskovec, Kleinberg, & Faloutsos 2005), physicists (e.g., Barabási & Frangos 2002; Newman 2003; Watts & Strogatz 1998), and statisticians (e.g., Handcock 2003; Snijders 2001). Network analysis now regularly uses information on large, longitudinal graphs representative of entire populations ($N > 2{,}000$) and entails information on multiple species and

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phenomena – from humans to dolphins, from neurons to power grids. In addition, network analyses now examine multiple types of relationships between entities in the same network – from friendships to marriage, from advice-giving to chain of command. Network analysis also continues to harness advances in software, computational power, and analytical methods to encompass even more expansive networks, such as social media interactions with even millions of observations, and to look at different ways that people relate through texts, shared activities or identities, or memberships in groups and organizations. Network studies are also going deeper into individuals' understandings of relationships through their perceptions of their own and others' relations. Going beyond descriptive accounts, today, a variety of structural hypotheses can be tested, and issues of causation can be explored in the context of networks. Moreover, decision processes (and algorithmic models thereof) are becoming central to our understanding of network formation (Jackson 2003, 2008). We further discuss some of these frontiers in our concluding chapter (Chapter 16).

In sum, the history of network analysis has been marked by steady conceptual, empirical, and methodological expansion, and by a cross-disciplinary focus on relational phenomena. However, in spite of the cross-disciplinary focus, the field lacks clear integration of the many theories and analytical methods now available to scholars. Network analysis is a pastiche of methods that span different software implementations and different disciplinary views, with no clear unifying perspective. Nearly every textbook on network analysis has been written for field-specific audiences by methodologists or authors using a field-specific set of tools and software packages. Moreover, there is a lack of awareness across fields currently engaging in network research – exemplified in particular by the tendency of hard scientists to overlook prior work in the social sciences only to "rediscover" what social scientists learned long ago. Physicists like Duncan Watts missed Granovetter's notion of bridges and weak ties in his concept of small worlds; the concept of Google's PageRank™ rests on the same metric as Bonacich's notion of power centrality and the Friedkin–Johnsen centrality measure (see Friedkin & Johnsen 2014); and even high-profile publications like *Scientific American* reproduce findings that social scientists identified decades prior (Paulos 2011; cf. Feld 1991).

We see the need for a more integrative approach to network analysis. The potential for less redundant, more fruitful collaborations is possible if researchers can integrate what increasingly appears to be a transdisciplinary perspective distinct from other scientific views. In our view, this integration requires recognition of network analysis's social scientific origins in theory and core empirical questions, and how these remain relevant to present research agendas and methods.

1.4 SOCIAL NETWORK ANALYSIS AS A FIELD-INTEGRATIVE FORCE

To clarify our motivations in writing this book, we begin with a clear statement of how we see the aims of network analysis. In our view, network analysis aims

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to characterize the pattern of transactions and relationships nested in the natural and social world and to examine both their antecedents and consequences. It entails understanding how associations form larger patterns and arrangements and how those relationships shift over time. Network analysis is grounded in systematic and purposeful data collection and analysis strategies, relies heavily on the use of graphic visualizations, and employs a far-reaching set of mathematical and computational models. Network analysis also encompasses efforts to understand how deeper structural principles shape relationships and how the configurations of relationships influence phenomena of interest, such as actor behaviors and attitudes.

The brief history we have related illustrates several clear divides in the growth of a transdisciplinary perspective of network analysis. The earliest period (from the 1930s to the 1950s) was denoted by mostly theoretical and qualitative research exploring basic concepts and relating them to social theory. The second period (from the 1960s to the 1980s) saw the emergence of a set of metrics and methods further elaborating network properties and their variation. Most recently (since the 1990s), a period of massive increases in scale and computing power has enabled network comparisons and hypothesis testing about network formation to a degree not previously possible. Each age has brought shifts in the type of scholar leading the charge – from theorist, to exploratory social scientist, to hypothesis-testing physical scientists and engineers – and a disconnect across what was learned in one era after the next.

In this text, we propose to integrate these views and to center the development of network concepts and methods around core questions of network structure and its formation. The key, we believe, is to tightly couple the methodological treatment with substantive questions that a researcher may hope to answer empirically. Such an approach is particularly important given the increasing availability of network data. Social networks and network thinking are more ubiquitous than ever because of the use of networking platforms like LinkedIn, Facebook, Instagram, and Twitter. Social networks are present in the endless number of forums, conversational threads, and streams of comments found on websites, online courses, and listservs. Companies, too, are awash with digital records and transactional data represented in streaming relational databases that they are not sure how to use.

In short, today research is experiencing a new empirical watershed of digitized communications, which has hastened the emergence of *methodological transactionalism* – that is, the capacity to empirically capture and theoretically explain *interactions*, which are frequently the traces of relations, observed at various levels from face-to-face encounters to global flows of goods (Kitts & Quintane 2020; McFarland, Diehl, & Rawlings 2011). For much of the mid- to late twentieth century, the individual was a practical, reified source of information collected through surveys. Today, that information comes from digitized social transactions, and the streaming of relational information has made the individual merely a point buffeted along within rivers of transactional data.

So what is not to love for a social researcher? The trouble with much of the contemporary research on networks is that novices learn a single method, acquire a network data set (e.g., Twitter), and then without reflection apply the method to the data. This approach, like using a hammer so ubiquitously that everything comes to look like a nail, can often lead to poorly executed or inappropriate analyses given the data and research questions. Most methods appeal to a particular question or class of questions and therefore do not apply to every problem. In addition, the problem that a method was meant to address can often have little relevance to the focal phenomenon in question. For example, in studying who retweets whom, a researcher may pick up a few network ideas of social influence that were based on how individuals in small groups experience conformity pressures, using these to "make sense" of thousands of tweets among total strangers for whom the original network conformity pressures have almost no chance of actually operating. Thus, there is often a gap between technical capabilities and more conceptual understandings. And many treatments of current methods in network analysis have only widened this gap by presenting methods with little conceptual context.

In contrast, this volume offers an integrative approach to network analysis that will be useful for filling the gap between methodological sophistication and theoretical and empirical purpose. For those scholars lacking technical capabilities, this volume can fill the technical gap concerning *how* to do structural analysis while helping to build and develop their theoretical agendas. For those scholars with more advanced methodological skills, our approach can help to bring these technical capabilities to bear on the broader theoretical landscape to elucidate *how*, *when*, and *why* these methods are so vital. We will demonstrate repeatedly that what students of network analysis often think is a methodological problem *is really a theory problem in disguise*.

Our main conviction is that *social* networks are the best possible bases for illustrating intuitive examples that bring together theory and practice and thus help integrate the transdisciplinary field of network analysis. All researchers, regardless of their discipline, can relate to the social world – for example, through the common experiences of attending schools and coming of age in high school or its equivalent. Most people wanting to learn about networks share a common set of understandings and experiences rooted in tangible, if not often surprising, social network phenomena. The same cannot be said of the many other types of networks – neural networks, gene networks, and so on – that are the bases for other fields. These are vitally important areas of research, but they cannot help unify the field of network scientists; only social networks can. Social networks are historically the origins of the field, and we believe that social scientists hold a key source of knowledge that can lead the field forward into a more fully integrated transdisciplinary future.[1]

[1] And, as we discuss in Chapter 12, sociology in particular holds a unique position as source of knowledge integration within the social sciences due to its central position and fairly weak paradigm that allows it to assimilate findings from numerous fields and disciplines.

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1.5 TWO BRIEF EXAMPLES

All networks comprise interdependent parts. But how do we illustrate and begin to analyze such interdependencies? A simple, yet powerful example of interdependence is easy to find in most American high schools and adolescents' onset to sexual encounters. Figure 1.2 shows a network of sexual encounters – in this case, within a single high school in the Midwestern United States in the 1990s. Each dot is a student, and the connections represent one or more sexual encounters over the year.

The structure matters. Individuals with the same number of partners vary in important ways in their position in the larger structure. Some individuals are indirectly connected to a large portion of the network, vulnerable to a sexually transmitted disease (STD) spreading through the large branch-like structure shown at the top of the image. Others are more isolated. Thus, individuals with the same number of partners (i.e., exhibiting the same behavior) may have very different risk profiles. If the researcher wants to understand how such a structure comes about (its etiology), how it affords students different opportunities for sexual partners, and the implications for the transmission of (for example) STDs, then the overall connectivity of the structure and each individual's position within that structure are of vital interest. The structure looks like a spanning tree because these youths mainly limited themselves to one or two

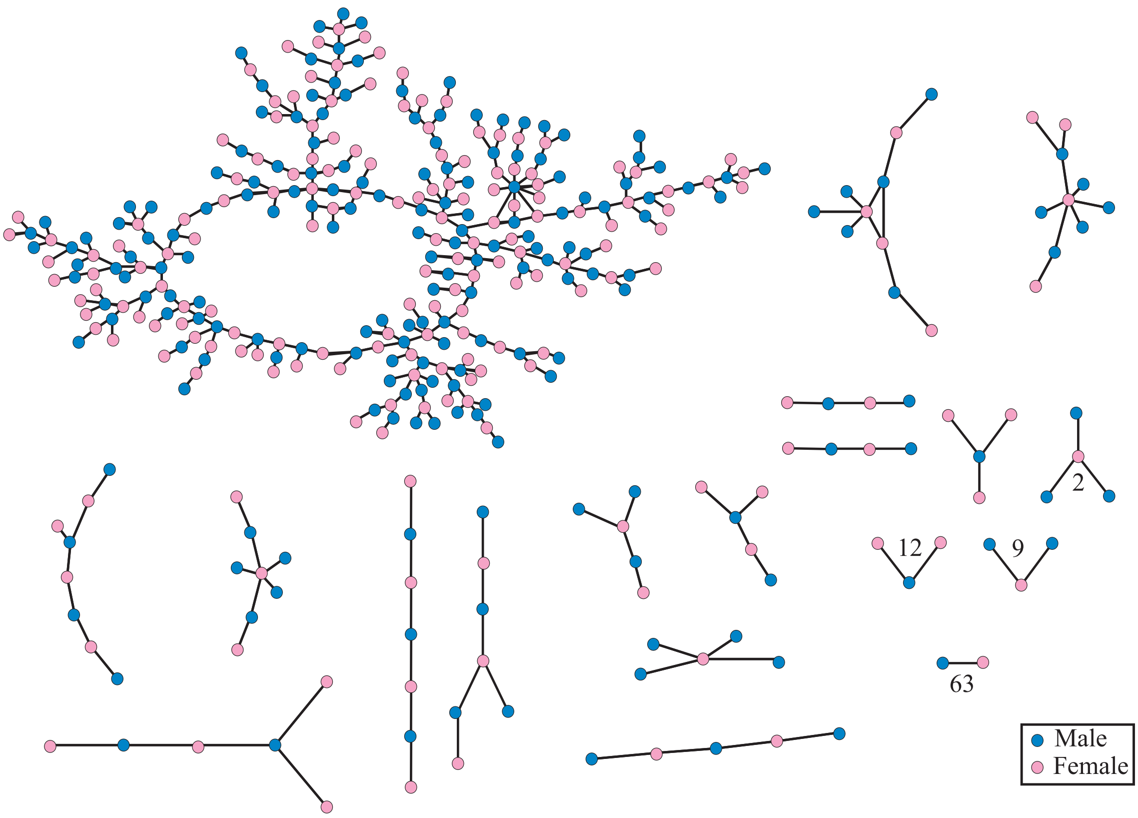

FIGURE 1.2 High school sexual relations network (Bearman, Moody, & Stovel 2004)

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romantic partners per year, mostly within the same school context. Some students more central to the network have greater access to other partners and are key players in the potential transmission of an STD within the school sex network. Clearly, taking these individuals and relationships out of context would lead to the omission of this vital information.

These structural principles can extend to other sorts of actors, such as countries. Figure 1.3 offers an illustration using three countries instead of people. The main question is whether Country C will attack Country B. From a network perspective, the answer depends not simply on the characteristics of Country C (e.g., its gross domestic product [GDP], military history, and party in power) but also on its relationship with Country A and the relationship between Country A and Country B. As shown in panel (b), Country C is in fact embedded in a larger system of relations: Country C is in a coalition with Country A, which in turn has attacked Country B. This means that Country C is allied with a country (A) that has gone to war with Country B. This may force Country C itself into a conflict with Country B (as an ally of Country A) even if B and C have no direct dispute with each other. In short, an enemy of a

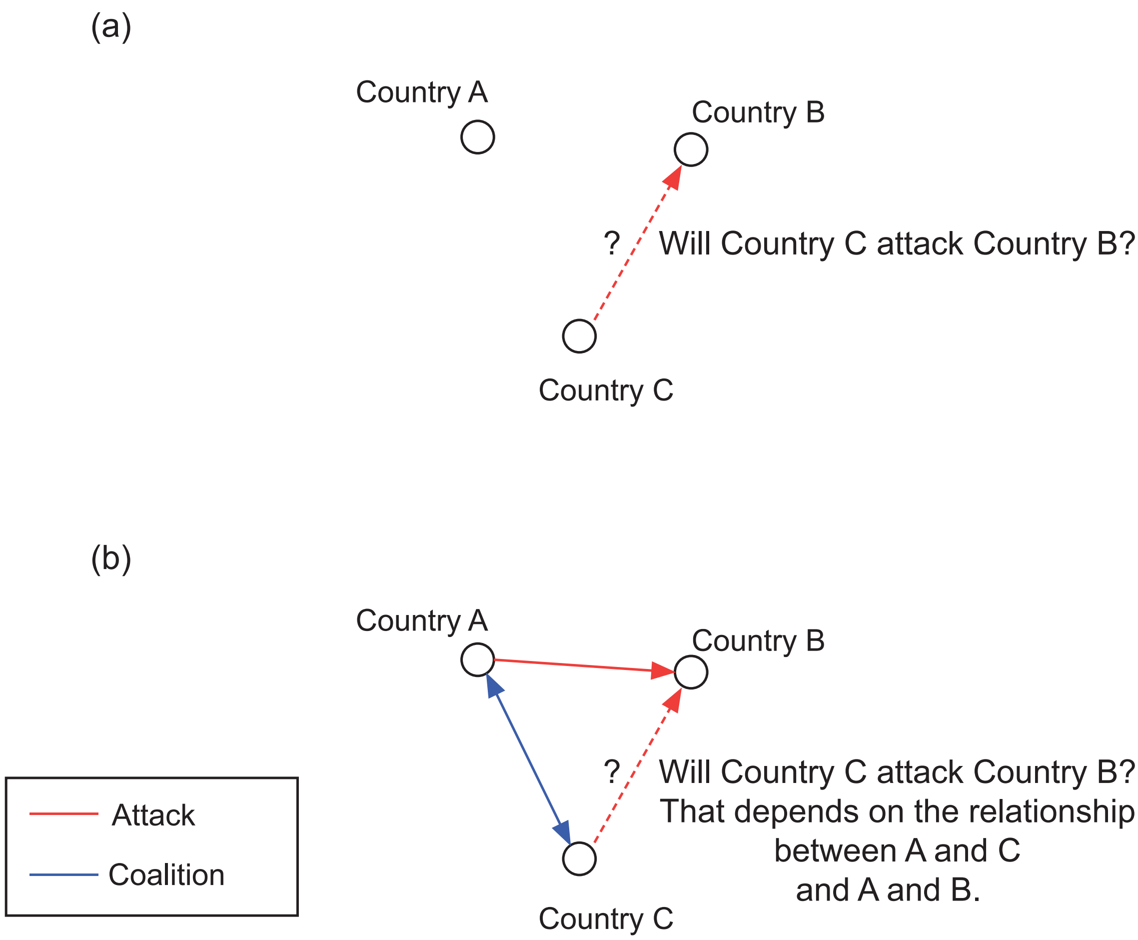

FIGURE 1.3 Structural forces in international relations

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friend is an enemy. The behavior of Country C would be difficult to explain if one considers each country in isolation, as in panel (a).

Hopefully, these brief examples and our discussion leading up to them has convinced you that structure matters and that social networks offer intuitive ways to begin to think about structures more broadly. But how to actually begin seeing and analyzing structures is no simple task and requires a set of tools.

1.6 SELECTING THE BEST TOOL FOR THE JOB

How should a researcher go about answering network-based questions? While there are many options, this book uses the R statistical programming language and platform to practically walk through the application of network analysis (R Core Team 2020).[2] We believe that R provides the best and most comprehensive set of tools, and becoming competent in this programming environment presents the fewest barriers for those with less coding experience. R is ideal, in part, because it avoids many of the drawbacks of other options, particularly those based on drop-down menus (i.e., point-and-click logic). Although these other packages often have a gently sloped learning curve, they make it difficult to custom-tailor the analysis to the data and research question under consideration. More importantly, they are not designed for replicability or extensibility. Accomplishing desired data transformations and routines to replicate analysis with these other packages often requires hacks involving a complicated give-and-take between a spreadsheet editor and other graphical user interface tools. This process is inefficient and error prone, and it can make diagnosing errors difficult. Moreover, these software packages are mostly stand-alone, closed-source applications, which means that building in additional functionality and creating methodological innovations are difficult, if not impossible.

In contrast, R has a vast array of powerful scripting functionality, exceptional visualization capabilities, and thousands of libraries to facilitate data management and statistical analysis. R excels in facilitating the development of new methods and approaches relative to other programming languages, while making it easy for an advanced R programmer to write interfaces to high-performance tools available in compiled languages, such as C and Java. It also interfaces with other environments (notably, Python), which is convenient for scientists and engineers who have already invested in those programming languages. Although no convention is a perfect solution, R has the added advantage of being shareware, both free to the user and open to improvements on existing techniques as well as the incorporation of techniques at the cutting

[2] For an excellent introduction to exploratory network analysis with the stand-alone program Pajek, we strongly recommend De Nooy, Mrvar, and Batagelj (2018).

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edge. R also allows access to myriad other statistical methods that many network researchers will likely want to draw on in their analyses.

Obviously, no single and perfect tool exists for performing network analysis. The tutorials we offer as accompaniments to the following chapters are meant to be adaptable to a number of research interests and to set the practical foundation for the conceptual material we present in respective chapters. We believe that having a common research tool such as R is also a basis for integrating the field as researchers across disciplines and for building a shared language and repository for generic structural analyses.

1.7 OUTLINE OF THE BOOK AND USER'S GUIDE

This book is the result of a collaboration among four social network scholars with distinct but overlapping areas of expertise. Rawlings has developed a number of ways to interrelate social structures with mental structures (e.g., attitudes, beliefs, tastes) using social network theories and methods. Smith has published work on social networks and health, methodological issues in network sampling and missing data, and has extensive experience in developing network methods in R. Both Moody and McFarland have published extensively on social networks, with particular strengths in building tools for better understanding dynamic social networks. Moody has additional strengths in social diffusion models and cohesion. McFarland has applied network methods extensively in educational and organizational settings. Together, the four of us have more than sixty combined years of experience teaching social network analysis at the undergraduate and graduate levels. We have sought to distill that collective experience into this volume.

The book can be many things for many different people. It is primarily offered as a research tutorial, offering students the opportunity to develop and answer questions that exemplary social network scholars ask when studying social phenomena. The book's orientation is to introduce methods of social network analysis in a theoretically grounded fashion; it does not cover mathematical modeling and simulation except when necessary. We hope to take the reader through every step necessary to answer core questions and to learn how to evaluate and interpret empirical results. The material can be tailored for more general or specific goals. For the network scholar who is already familiar with network theory and methods but wants to become more proficient at R, the research tutorials afford an opportunity to move firmly into this new programming environment in a way that is more theoretically grounded than many other texts. Instructors of graduate or undergraduate courses might rely on the book in its entirety or instead choose portions that are appropriate to cover conceptual and empirical applications or laboratory work as the course requires.

Each chapter contains several elements: (1) identify core research questions; (2) relate the influential texts and their concerns bearing on these questions;

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(3) postulate an appropriate plan of analysis; (4) choose the appropriate methods (and compare them); and (5) interpret the results, their quality, and how to present them as findings. The text offers concrete examples – many from contexts most readers have experienced firsthand (e.g., classrooms) – with real data. The tutorials are available on the web at: https://inarwhal.github.io/NetworkAnalysisR-book/.

We divide the book into three main sections. Part I focuses on structural thinking, introducing the main concepts of network analysis, identifying the key visual and mathematical abstractions that form its core, and discussing issues of data collection. Parts II and III cover two, often interrelated, analytic goals of network analysis. The first, presented in Part II, concerns using a number of *exploratory* techniques to see structures at various levels and degrees of abstraction. In particular, we help the reader develop connectionist and positional perspectives on network structures. Part III concerns making structural predictions using a variety of more dynamic, longitudinal, and *explanatory* models.

No textbook can cover every method, and we have made some painful but necessary omissions. We therefore end each chapter, except for the conclusion, with a short list of works that will expand on some of the core ideas developed in the chapter, either by building depth or extending to detailed areas that are beyond the scope of the chapter itself. The universe of works we could include is vast, so any such lists are necessarily incomplete and idiosyncratic, but we hope these serve as useful jumping-off points for readers. In addition, given the rapid development of the field, it is likely that new techniques are currently being developed that could surely join those presented in the final sections of Part III. We hope to include such exciting new work in future editions.

SUGGESTED FURTHER READING

Overviews and Introductions. The following works are recommended as general introductory works that cover history, theory, and applications.

Barabási, Albert-László. 2002. *Linked: The New Science of Networks*, Cambridge MA: Perseus Press. (Provides an interesting systems-science foil for social science approaches. Barabasi, Watts, and Newman are key figures in the late 1990s rise of "network science" as distinct from social network analysis. See also Watts 2003; Newman 2018.)

Butts, Carter T. 2009. "Revisiting the Foundations of Network Analysis." *Science* 325: 414. (A critical summary of the idea that all connected systems are "a network," and highlights the need to tailor approaches to the complexities of empirical settings.)

Easly, David, and Jon Kleinberg. 2010. *Networks, Crowds, and Markets: Reasoning about a Highly Connected World*. New York: Cambridge University Press. (An overview introduction with a focus on network science approaches to economic and financial questions.)

Freeman, Linton C. 2004. *The Development of Social Network Analysis: A Study in the Sociology of Science*. Vancouver, British Columbia, Canada: Empirical Press.

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(Provides a rich history of the development of social network analysis as a substantive discipline.)

Jackson, Matthew O. 2008. *Social and Economic Networks*. Stanford, CA: Stanford University Press. (Brings economic modeling/theory to networks.)

Kadushin, Charles. 2011. *Understanding Social Networks: Theories, Concepts and Findings*. New York: Oxford University Press. (A substantive introduction to structural theories of social life, with clear applications. The "ten master ideas" chapter, in particular, provides a succinct summary of why networks are fundamental to understanding social processes.)

Light, Ryan, and James Moody. 2020. *The Oxford Handbook of Social Networks*. New York: Oxford University Press. (A broad overview covering multiple contemporary topics by field experts.)

Lin, Nan. 2002. *Social Capital: A Theory of Social Structure and Action*. New York: Cambridge University Press. (Lin's work illustrates how social connections and social relations can be a key resource.)

Watts, Duncan J. 2003. *Six Degrees: The Science of a Connected Age*. New York, Norton. (A substantive introduction to the network science approach. See also Barabási 2002; Newman 2018.)

Wellman, Barry. 1988. "Structural Analysis: From Method and Metaphor to Theory and Substance." In *Social Structures: A Network Approach*, edited by Barry Wellman and S. D. Berkowitz. Cambridge: Cambridge University Press. (A nice introduction to the what and why of networks.)

Key Historical Works. The field has evolved over the last 100 years with touchstone works that are guideposts for much of what drives our contemporary understanding of network theory. This list is far from complete and chosen mainly to represent the substantive breadth of foundational approaches rather than completeness.

Baker, Wayne. 1984. "The Social Structure of a National Securities Market." *American Journal of Sociology* 89: 775–811. (An exemplar of how structural realities undermine pure market assumptions. A classic work linking networks to economic sociology.)

Coleman, James S. 1961. *The Adolescent Society*. New York: Free Press. (Demonstrated that adolescent networks and schools worked as largely self-contained social systems characterized by social networks. See also Hollingshead 1949.)

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PART I

THINKING STRUCTURALLY

Network analysis formalizes the study of relationships, and social network analysis pertains to relationships among individuals, groups, or other social entities. While an assortment of social network approaches exists, greater integration is needed to guide our efforts. In this section, we first outline our theoretical integration based on (1) describing social structures and (2) pursuing strategies for better understanding social structures within a causal framework. We then cover in more detail how network analysis formalizes the conceptualization, data collection, and visualization of relational data.

2

What Is Social Structure?

The primary aim of social network analysis is building and evaluating theories of social structure – that is, enduring patterns of human interaction and ways of thinking about and organizing human groups. The sheer complexity of social structure prevents encapsulation in any single model, and this complexity is compounded as we incorporate cultural beliefs and social expectations in addition to interactions. Networks link actors to one another in systems, raising tricky questions about the locus of control and activity, particularly regarding the extent to which people are active agents or passive puppets (to put it bluntly) of social structure. While acknowledging deep and ultimately unsettled issues in the field, we provide readers with an overarching though still evolving theoretical account of social structure that can guide both inductive and deductive social network research and allow plug-in points for different perspectives on agency, culture, and constraint.

2.1 METATHEORY PROLOGUE: FROM REALITY TO REPRESENTATION

Research often begins with imagination. One might have an interesting idea about how things work in the world that goes beyond current assumptions. One is sometimes right, but one is often wrong. And realizing when and how one is right or wrong often requires insights from others working on similar problems. Research therefore builds on the wisdom and cumulative efforts of countless individuals because reality is often hidden and far too complex for any one of us to see on our own. In fact, left to our own mental devices, we are likely to hold onto bad ideas in the hope that our cherished imagination of something interesting will eventually prove true. Thus, research can be defined as the *collective* process of rendering some aspect of reality into data and then processing those data into useful information (i.e., knowledge).

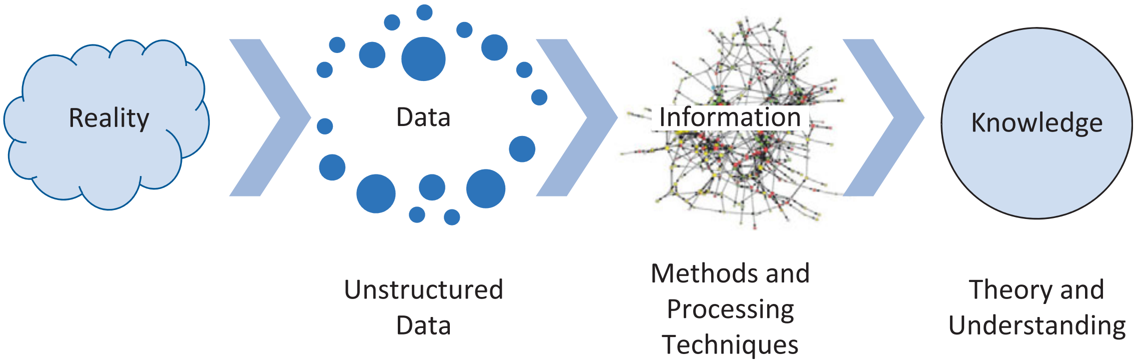

FIGURE 2.1 Schematic of rendering reality into knowledge

Theory is present at all stages of the research process. When we observe phenomena, we do so through the lens of established research traditions and paradigms of thought. These traditions entail their own language and ontology as well as distinct epistemology or means of sense-making and inference. These theories act like frameworks of interpretation, leading researchers to observe certain aspects of a phenomenon over others. Even highly inductive research is guided by sensitizing concepts (i.e., theoretically informed ideas) that select on certain social aspects of reality over others. Likewise, theoretical perspectives guide the choice of certain data collection tools. For example, as we discussed in the previous chapter, methodological individualism led researchers to collect individual-level responses to survey questions and interviews, and these became state-of-the-art tools for data collection in the 1940s. Social network theory, however, shifts the focus to relations, and it is aided by a different set of data collection tools that nudge researchers closer to the reality of the interdependencies among objects. The focus on relationships guides the researcher to find data in transactions, affiliations, links, and comparisons *across* objects. In this way, theories lead us to discover and collect certain forms of data from social reality.

Data are merely traces and representations of social reality guided by theory. Transforming these data into useful information is also guided by theory. For example, in social psychology, transforming self-reported feelings into a measure of self-esteem presumes a theory of self-esteem and some latent mental mechanism that enables those sentiments to work together. The same holds for social network theory. It sensitizes researchers first to collecting data on the interrelation of objects and then to identifying processes and patterns in those relations. These processes and patterns become information that is interrelated into larger, holistic understandings about social structures. For example, the social network theory linking social proximity to feelings of intimacy entails collecting reports on the strength of personal ties and measuring the social distances between persons, and then ascertaining the incidence of stronger interpersonal affect with social proximity. As depicted in Figure 2.1, researchers use theories to identify the observations or data they collect, the ideational constructs and measures derived from them, and how the

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relationships contained in these pieces of information are useful for some broader understanding.

Some social realities and phenomena are more tractable with certain kinds of data and seem best suited to particular constructs and measures, which in turn are best suited to certain methods and theories. Of course, the reverse can be true: one's preconceived knowledge and theories determine one's choice of methods, which can determine the constructs one builds and in turn shape the data one collects and the social reality one observes. The point is to not be naïve about this process of translation but instead to enter it with eyes wide open to make the most reasonable decisions possible at each juncture. Consider whether you are biased in perspective and weigh alternative data, measures, methods, and explanations. Consider the phenomenon and whether you are representing it fairly or whether your training guides what you see (Hollis 1994).

We argue that a social network approach often renders a more accurate representation of social reality than other social scientific approaches because it recognizes the fundamentally interdependent nature of social life: to be social means to be connected to others (Blau 1977). This is not to say that one method supplants all; rather, a reflexive, smart analyst must step back and ask, What representation of this reality is most suited? What constructs and alternative forms are best suited to these data? And what methods and explanations – again, including alternatives – are most relevant? This approach also highlights a core distinction between building understanding and mere prediction (see Turco & Zuckerman 2017; Watts 2014). Prediction alone is insufficient for most social science questions; the researcher simultaneously seeks description and explanation – that is, understanding. Understanding centers on linking theory, data, and prediction in a way that recognizes their mutual information. The purpose of this chapter is therefore to begin to orient the reader's imagination toward a view of social structure that will help guide these research efforts.

2.2 AN EXAMPLE: SOCIAL STRUCTURE IN A CLASSROOM

Social network scholars seek to identify and understand patterns in relations between actors. This patterning is called *social structure* when it is stable enough to be reproduced and recognized via some ostensive label, such as "friendship," "marriage," or "trade." Social structure is at once a pattern of interactions in the world and a set of schemas (simplified mental abstractions) for recognizing and enacting such interactions.

Imagine someone who has never seen a classroom. How would they make sense of its complex set of interactions? A bell rings; individuals (generally younger ones) scurry toward desks oriented in the same direction, and often choose seats near the same people each time; and at last, a single individual (generally an older one) enters and moves to the front of the class to speak, as

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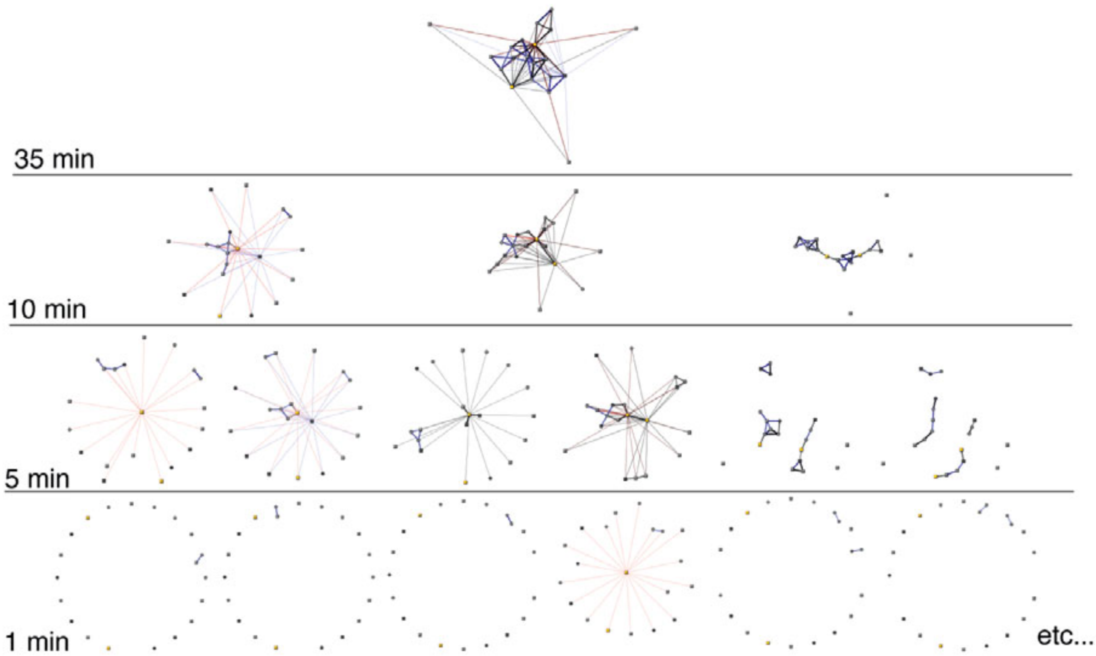

FIGURE 2.2 Networks from slices of interactions using classroom observation data (Bender-deMoll & McFarland 2006)

those who are seated start to settle. As Berger and Luckmann (1966) describe it, the social construction of reality involves individuals seamlessly engaging in a process of externalization (relating), objectification (naming), and internalization (learning) of structures. The social structure of the classroom is therefore one that exists in schematic form in the minds of the individuals via roles and relations. But the performance of these roles and relations is contingent on how individuals coordinate their interactions to more or less successfully cue and enact those schemas. The result might be a loose performance of roles with much humor in one class and a strict script with lots of private grumbling in another. Both settings share a structural semblance of being classrooms in terms of how people interact, but they vary in the specific instantiations.

But how can one begin to see and understand such a structure? Learning how to do social network analysis is first and foremost about learning to have theoretically guided hunches and the hunger to solve puzzles about social structure – that is, an informed social-structural imagination. By the time of data collection, the researcher has already engaged in some conceptualization of social structure. Consider the networks in Figure 2.2, all taken from the same single class in a high school (Bender-deMoll & McFarland 2006). Ties depicted in black are task-related interactions, blue ties are sociable interactions, and red ones are conflictual ones. Circles are girls, and squares are boys. Yellow squares are male teachers, and the other nodes are students. Notably, the class has two teachers: one lead teacher and a volunteer.

Structure begins to emerge as interactions aggregate over larger chunks of time. At one-minute intervals, only sequences of dyads are observed.

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Aggregating these interactions over a wider time span of five-minute intervals reveals patterned relations reflective of activity structures and the speaker roles adopted to accomplish them. For example, on the left side of the figure, consistent star structures are reflective of a teacher repeatedly emitting broadcasts, small blue dyads reflect side socializing, and black dyads reflect clarification efforts. Moving left to right in the figure, some students grow closer to the teacher. These are instances of recitation in which a teacher uses broadcasts to demonstrate and explain things but then calls on particular students for question–answer sequences. Moving further to the right, some students assume a broadcasting position, reflecting their broadcasting information to the class, such as when they report out or give a speech. And to the far right of the figure are clustered social and task interactions reflective of free time and group work activities. When interactions are aggregated to ten minutes, one begins to see general activity structures, not just momentary ones and their instantiation. The classroom is characterized by a contentious lecture (with a student challenger), a "group work" instructional format, and then a collaborative group structure. If one aggregates to a typical class period of thirty-five minutes, distinct activities are melded into one graph revealing the daily social structure, as an abstraction, for a classroom: the teacher is in the center, most students and another volunteer teacher are clustered around the teacher (some via social and others via task interactions), and some peers are at the periphery.

Such an aggregated image of a micro-level social structure reflects what the teacher might regard as an *interactome*, a term used in biology to refer to the entire set of molecular interactions in a particular cell. Here, one can think of an interactome as the manifest social structure and the typical associations among setting members. Extending this notion further afield to many classrooms, one can abstract out social roles and styles, such as central teachers, social and task-oriented students, and isolates. These schemas can be cued in interaction, framing expectations, and influence ensuing social interaction. They can also fail to come about, reflecting the processual and contingent nature of social structure. None of this structure would be visible from a random survey of students and teachers.

Still, such snapshots of social structure leave open questions – for example, what drives this structuring at each level? This question is much more theoretical than methodological. A variety of factors can establish opportunities and models for the observed interactions. It may be that persistent interactions follow expectations of preexisting friendships and formal roles (teacher–student), differentiated and developed styles of interaction (e.g., "nerds" in tasks, "jokers" in social antics), or the opportunities afforded by proximate seating or shared histories (other classes and clubs), attributes (boys vs. girls), and attitudes (liking of class). In prior forms of structuralism, researchers relied on the logic of *functionalism* to explain observed structures – that is, they argued that resultant social structure served the deeply held social purposes of their members (generally some need for social stability and group cohesion) and that the structure evolved to fulfill these purposes in presumably optimal ways. Sociology as a discipline and social network research as a subfield have largely rejected structural-functionalist explanations as overly speculative and untestable, leaving us to account for structures in more falsifiable ways involving both endogenous and exogenous forces, which we will cover in detail in this book.

Network analysts are interested in studying the *process of structuring and how structural variation arises*. But how does one gather and analyze suitable data on such a complex and fluid reality? The answer is aptly illustrated by the parable of the three blind men and the elephant: while each man feeling a different part of the elephant believes it to be a different object (the trunk a snake, the leg a tree, the ear a fan), the men are able to communicate and put their stories together to form a more developed representation of social reality. Knowing something about social structure entails thousands of partial truths discovered by researchers making strategic trade-offs in targeting their efforts. But, once again, when researchers studying the same reality communicate and learn from one another, they get closer to a shared, replicable, and sustained notion of reality.

This book is, in part, an attempt to help consolidate these efforts to better guide the strategic directions researchers take. Any attempt to draw a picture of the current understanding of the "elephant" of social structure is bound to be both complicated and incomplete. Why would one need social network analysis if social structures were so easily seen? In the following, we provide a heuristic for seeing social structures and broad research agendas for making structural predictions. In the remaining chapters of Part I, we outline the basic tools for gathering and visualizing relational data. In Parts II and III, we outline in greater detail the inductive and predictive methods that are frequently involved in the rendering of these data into knowledge.

2.3 SEEING SOCIAL STRUCTURE

The core motivation of social network analysis from its inception was addressing the question, If one could see social structure, what would it look like? Although no single image can capture social structure, we revisit a classic attempt by Hinde (1976), reproduced and adapted in Figure 2.3, which shows several important aspects of social structure and can help focus future efforts by shaping research questions and research designs. Hinde was a primate ethologist (Jane Goodall's advisor, in fact). As such, he spent a lot of time observing primate interactions and trying to identify generalizable patterns and potential mechanisms shaping primate social structures. The figure is adapted from his effort to encapsulate the reality of social structure as both interactions among individuals and schemas or types of relations and mechanisms, which in turn compound into groups comprising different types of positions among individuals.

Although Hinde's model concerned troops of chimpanzees, he was able to abstract to more generalizable features of social structures. We adapt Hinde's

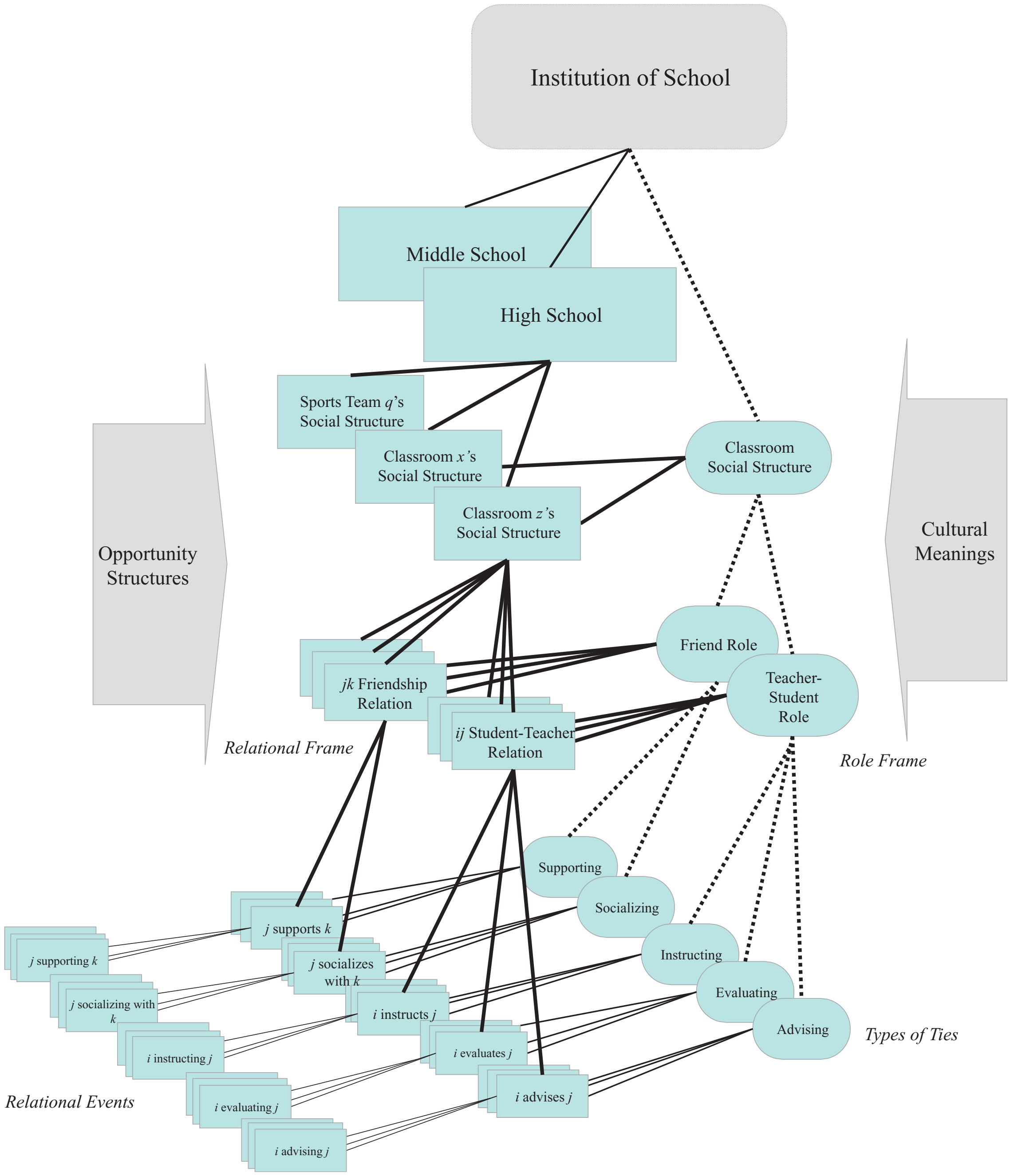

FIGURE 2.3 Schematic of social structure (adapted from Hinde 1976)

view to apply to the social structure within a high school classroom rather than among a troop of apes (although, technically speaking, a classroom is composed of a group of apes). For our purposes, the figure captures two key aspects of rendering any social structure into knowledge: (1) the conceptual degree of abstraction and (2) the empirical level of abstraction. Moving from the left to the right, one engages in a greater amount of conceptual abstraction – that is,

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from concrete instances of individuals interacting in particular ways (*relational events*) to abstract schemas of that interaction (*types of social ties*). Moving from bottom to top, one compounds abstractions to higher-order levels of structure – that is, from micro-interactions to role relations to groups to macro-institutions.

We use Hinde's image as the basis for depicting the social structure of a high school. Here, one sees concrete relational events between individuals occurring in the school as the base unit on the lower-left side of the figure. At the level of concrete interactions, one sees relational events specific to moments where *i* advises, instructs, and evaluates *j*. These relational events share a semblance and point to the ways that *i* and *j* relate with each other – that is, more general ways that *i* advises, instructs, and evaluates *j*. Moving to greater abstraction, one can see that these relation-specific instantiations of tie types resemble one another across pairs of individuals and reflect different types of ties (advising, instructing, and evaluating). Each move to the right entails further generalization and abstraction.

Moving upward in the figure, types of *ij* ties can come to reflect a general *ij* relation – say, how Craig and Jim usually relate to one another. Relation-specific types of ties combine in a patterned way to characterize other relations and their expectations – say, how Jim and Dan usually relate. These built-up histories for specific relationships establish relational expectations or *relational frames*. If we consider the similarities across these relationships, we start to see more general social roles like that of friends or teacher–student roles. Roles emerge from patterns of consistent social actions originating from and pointing toward actors (Nadel 1957). These patterns of consistent role relations often stem from following and conforming to institutionalized *role frames*.

To this point, we have drawn on Hinde's depiction of the most elemental unit of relational events and, through recognition of events' shared resemblance, identified types of ties. We have then aggregated these types of ties across particular pairs of individuals to characterize their particular relationships. And from there, a recognition of shared resemblance across particular relationships, has emerged general role relations. From here, we can aggregate again into larger configurations of social structure, such as the structure of a particular group or type of group and a society and even a type of society. If we observe and interrelate the set of relations within a context such as a specific classroom (say, classroom *x* or *z*), we come to see the social structure of those social networks. At an even higher level of abstraction – having observed many different classrooms with varying ages, compositions, seating, and activities – we may come to see classrooms sharing another family resemblance, mostly entailing centralized teachers in task types of interactions with students and students clustering in friendships and social types of interactions with one another.

Moving out of the classroom, but remaining in the same building, one then observes other individuals: in offices sorting papers, in kitchens making food, in sterile rooms taking sick adolescents' temperatures, and so on. Looking outside the window, perhaps one sees a sports team wearing uniforms that bear the name on the school building and engaging in activities involving an entirely separate set of agonistic interactions. All of these could be appended to Figure 2.3 as separate descending branches with their own relational events, relationships, roles, and role structures, as well as their accompanying schemas of expectation. Finally, one might then fly across the world, walk into a village building, and see children and adults arranged and interacting in similar ways to what one observed in the United States. At the highest level of abstraction, one may begin to see an institution: the supra-level of organization that reflects the deep cultural structure of school itself.

Rendering knowledge from snapshots of relating requires theory and a developed understanding of social structuration processes. Social structures are at once interactional, relational, and institutional requiring researchers to make choices about when, where, how, and why they gather data and render reality into knowledge. Otherwise, researchers may observe the same social reality of a classroom but be attuned to very different processes shaping that social structure. In what follows, we offer some of the many ways that social network researchers begin to make sense of social realities and usefully represent them.

2.3.1 Seeing Structure in Interactions

Once again, in contrast to much of sociology, which draws on reifications of social structures in the form of self-reports or records of enduring patterns of activity, social network analysis has been said to extend from an "anti-categorical imperative" (Emirbayer & Goodwin 1994) – that is, it seeks to move beyond the use of prescribed categories such as age, party membership, race, class, or gender as sufficient explanations of what motivates thought and action. Instead, social network analysis seeks to examine individuals' attitudes, beliefs, and behaviors through the lens of structures of interactions and relations (of which the aforementioned categories are simplified proxies or relational schemas). Beginning at the most fundamental level of individuals interacting and relating with one another, some scholars are interested in how face-to-face and moment-by-moment data can reveal structure, as well as how actors are motivated in some way to use that structure as a resource for pursuing various interests. Microsociology and scholars studying talk-in-interaction have repeatedly shown how social reality is an ongoing accomplishment in the form and content of interaction (e.g., Schegloff 2007). Thus, some network researchers train their eye on the lower-left quadrant of Hinde's image, rendering networks through actual moments of talk and interaction.

At the level of ongoing activity, social structure can be driven by *interaction moves* – that is, how individuals refer to themselves and others, as well as to the roles and rules within institutions. Interactions both form and reveal relationships: piggybacking statements and reciprocal turns may form friendships, changing topics and usurping turns may generate animosity, and commencing sequences may generate status (Gibson 2012; Heritage & Raymond 2005; McFarland, Jurafsky, & Rawlings 2013). For other researchers, preexisting relationships guide interactions. A classroom, for example, is composed of particular friendships, proximate seating, and specific activity structures that bring persons into relation – all of which motivate certain forms of interaction (e.g., reciprocation) and establish social expectations. These relational arrangements can give a class its particular style and ethos. For yet other scholars, institutionalized roles and norms define expectations, determining what relations and activities are salient, which in turn defines what types of interactions individuals are likely to employ in a given setting. Of course, these levels of concern and reference need not align in reality: individuals may adopt interaction moves that run counter to relations and institutional roles, such as when observing classroom resistance. And such mismatches of preexisting norms and ongoing behaviors can lead to gradual and long-term changes in social structures (McFarland 2001).

2.3.2 Seeing Structure in Relationships

Much of social network analysis is concerned with enduring relations rather than surface-level moments, although (as we will discuss in several places in this book) the techniques and approaches are amenable to modeling real-time interaction data (Moody, McFarland, & Bender-deMoll 2005). The interpolation from fluid to structural, however, is no small task. One knows a relationship when one sees it, but how the researcher operationalizes a relationship is already full of consequence. One generally has to infer a relationship from interactions, but how much of an actor's mental model of what is going on in a given interaction is needed? What types of indicators best capture the existence of a tie? A reflection on what relationships are and what it means to relate (in common-sense terms) quickly reveals relationships to be perceived, enacted, emotionally experienced, and socially confirmed; relationships have *cognitive*, *behavioral, affective*, and *interpersonal* aspects (see Bandelj 2012; Fuhse 2015).

- The *cognitive* aspect of a relationship is the relational *framework* that one perceives the relationship to be in or wants to form. For example, many relationships are perceived to be friendships, acquaintanceships, or dating relations, and each entails different social expectations and obligations.

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- The *behavioral* aspect of a relationship is observed in *social signals*, or the actions and expressions we exhibit in interactions. It is seldom enough for dyads to believe they are friends; they must also interact in ways that support their partners and affirm the friendship.
- Relationships also have an *affective* aspect, which concerns feelings or *sentiments*. In the case of strong ties like friendship or romance, participants feel strong positive sentiments toward one another and value the relationship. This ensures that relational obligations will be preserved even when the alter is absent.
- Last, a relationship rests on *interpersonal* aspects of *agreement* across persons as to what relational framings are believed, what enactments are observed, and what interpersonal sentiments are held. When cognitive, behavioral, and affective aspects of ties are reciprocated and socially consistent across persons in a wider context, the relationship is no longer privately held, but is socially confirmed.

Take, for example, relationships in a high school. Youth may believe and report they are friends (as opposed to dating), affirm the relationship by showing support and spending time together, actually like one another and value their relationship, and come to have a mutual sense of these relational beliefs, observations, and sentiments. However, adolescence is rife with change, and all too often youth are uncertain about their relationships. Consider the following interaction, taken from written notes passed between students in a high school classroom (a relic of a pre–cell phone era):

[Meg]

Do you know Steven Johanson? Alot of people think he's a geek, I guess. But he likes me and he's so nice. We talk on the phone alot and I went over to his house last night. Nothin' happened but he is really nice and his family is nice, and he has a huge house and a pool. (Asshole! J/K) His sister is pretty, she doesn't look 12 ½. She looks like she should be in 9th grade. A lot of people told me not to worry about what other people think. I asked him to TWIRP ["The Woman Is Required to Pay"-Dance] (kind of). I still have to figure out what's happening. I don't know what we'd do or where we'd go or who with. You're probably thinking I'm crazy to go out with Steven, I hope you don't think he's a big nerd cuz I know he's not super popular or anything, but not alot of people really know him, and once you get to know him, he's super nice. Anyway, better go. W/B very soon.

[Laura]

I know Steven pretty well, he's a great guy. I think it would be awesome if you 2 went to TWIRP. He is just shy, not a big nerd, Sarah [his sister] is really pretty, we play tennis together.

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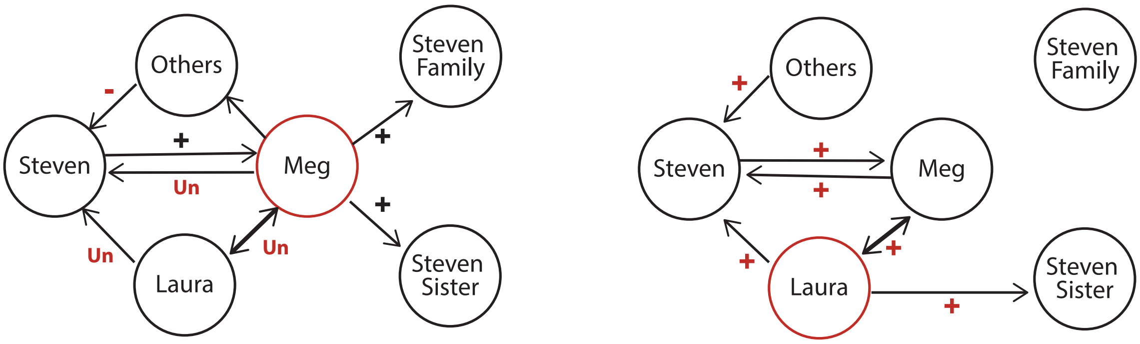

FIGURE 2.4 Social structure constructed in notes passed between two high school students (McFarland & Wolff 2022)

The conversation is deeply concerned with social structure. Figure 2.4 depicts the social structure and various relational uncertainties being negotiated through this written exchange (see McFarland & Wolff 2022). Meg reports that Steven likes her and that he is so nice – a reciprocal sentiment. She also reports that they are going to a dance together – a behavioral signal of them dating. While she does not frame the relationship as one of dating, it is evident that is the direction the relationship is heading. What is mostly at issue in the exchange is Meg's uncertainty as to whether this relationship with Steven will create imbalance and relational uncertainties for her friendships with Laura and others. Meg is mainly seeking social affirmation of her interest in Steven and her decision to invite him to a dance. She seeks Laura's input on the state of several dyadic and triadic relational uncertainties, particularly Meg's feelings toward Steven (and his possible status as "a nerd") and Laura's feelings toward Meg (and her possible status as "crazy"). Laura's response quickly addresses these uncertainties and introduces reciprocity and balance to the projected network of interpersonal sentiments. Laura indicates that Steven is "a great guy" and that Meg is definitely not crazy. Laura affirms both her positive feelings toward Meg and Meg's uncertain feelings toward Steven. Agreement is even had regarding Steven's sister when Laura notes her beauty and prior contact. Meg and Laura are engaged in finding *interpersonal* agreement about the *affective*, *behavioral*, and *cognitive* aspects of their social ties. This sort of communication about relationships can be seen as relational work more broadly (McLean 2007).

As one moves from micro-level interactions of relations to more stable, macrolevel relations, one looks to aggregated data on interactions: to individuals' enduring relational perceptions, feelings, and behaviors. Perhaps one loses something in this shift upward in Hinde's image if one does not account for deeper cognitive and affective dimensions of relating. If one infers relations by observing repeated interactions or simply asking individuals who they "like" or who are their "friends" without greater understanding of what such terms *mean* to the individuals and how these meanings may actually differ in systematic ways among subgroups (Gondal 2022), one may see a strange or even

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biased social structure (Kitts & Leal 2021). It is therefore vital to account for the right side of Hinde's depiction, in which social structures exist as those categories and meanings that shape (but do not determine) how actors interact and relate to one another.

2.3.3 Seeing Structure in Schemas and Meanings

As Figure 2.3 illustrates, an inherent connection exists between social structure and schemas – that is, how people understand who they are in relation to one another in a given social situation. Take, for example, the game of baseball. There, the players know their positions and the rules of the game. These institutionalized roles and codes of conduct, in fact, *are* the game of baseball; the individuals on the field, as Shakespeare would say, are "merely players." Thus, as we will discuss in Chapter 12, network research blends into cultural sociology in its search to better understand how meanings and actions interrelate (McLean 2016). When and how such categories matter in shaping networks and when networks shape those categories are unsettled questions that represent an area of active inquiry.

Bracketing out the causal link between structure and culture for the moment, we can draw attention to more inductive and interpretive work that attempts to map meanings and schemas within various institutional contexts. If actors in networks are merely players in any number of more or less institutionalized games, then where does one find the "rule books" for these games? For some, the answer is in the heads of participants, pointing to the importance of questionnaire and interview data to supplement observations or reported relations (e.g., Crossley 2010). For others, the discourse and symbols surrounding an institutional domain – from how the chairs are organized in the classroom to the textbooks – offer insights into the schemas and meanings organizing and making sense of actions. Thus, researchers may turn to the material and nonmaterial culture that helps organize an interactional setting or (at a higher level of aggregation) an organizational field of activity in order to better see how underlying schemas, meanings, scripts, and so on provide the basis for roles and thereby lend meaning to interactions.

How deep to go in looking for cultural codes is a matter of debate. Some researchers have drawn on a long tradition of structuralism in social anthropology that focused on the deep structures of culture that shape the human mind. Consider the example of Bourdieu's discussion of the Kabyle house as a "structuring structure" (Bourdieu 1984: 170). Bourdieu depicts the layout and orientation of the typical house of this North African ethnic group through a series of cultural oppositions: wet/dry, male/female, human/animal, sun/moon, and so on. How one enters and exits the house and how activities and mundane goods are organized in this space are part of a social structure composed of familial and gender role relations that are enacted and given meaning through the *homologies* that exist within that space – that is, through differences in the social structure that are embedded within layers of symbolic differences in other domains (physical space, cosmology, objects, and the animal world). Other scholars link social structure with deeply ingrained notions of purity, danger, the sacred, and the profane.

Cultural accounts of structure get into very murky waters when trying to align with theories of evolutionary psychology or functionalism, which explain culture as serving the purpose of social cohesion and which, as we have already discussed, sociologists have largely rejected in favor of more "middle range" theories.[1] Some approaches, such as that of Douglas (1970), went very deep indeed and reduced the structure-culture connection to a two-by-two table with social cohesion as the primary explanation. Rather than seeing culture as a force for creating normative cohesion in society, sociologists have viewed culture as a type of resource for achieving various practical goals and making sense of a flowing, often indeterminate reality. Other scholars have demonstrated that shared meanings are not essential for social structures to enable and constrain our actions. For example, Bearman (1997) showed that an entire macro-level system of generalized exchange in brides among Aboriginal Australians does not arise from any articulated rule and shared cultural understanding so much as from each local group knowing what feels right and which arrangements will not cause conflict for others in their local community.

As network scholars have been theorizing for some time now, agency within the context of structure and culture is difficult to discern (Emirbayer & Goodwin 1994; Emirbayer & Mische 1998). We cannot possibly do justice to this work here, but we believe it is important to emphasize that social network approaches are nondeterminative with respect to how agents will negotiate social structure and culture. As we will discuss in somewhat more detail in the next section, the ability to violate cultural rules, innovate, and improvise has the potential to transform existing structures. In other words, social reality is complex, and the researcher needs to keep thinking structurally if they are to use social network methods to realize this reality more fully. In the next section, we describe four perspectives researchers can adopt when performing structural analysis that make doing so more tractable by narrowing the focus.

2.4 SOCIAL STRUCTURE AND PREDICTION

Accurately observing and describing social structure is the first major contribution of social network analysis. The second, which has come more and more to the fore as computational power has grown, is asking where networks come

[1] A milder form of structural-functionalism is making a comeback of sorts using more evolutionary perspectives to explain elementary forms of sociality (e.g., Christakis 2019; Haidt 2012; Henrich 2016; Ridgeway 2019).

from and what they might be doing in the world. In the image by Hinde shown earlier, such theorized large-scale causes of social structure are shown as arrows outside the structure. These causes often reflect deeply held norms, orientations, and latent constructs that are deeply ingrained (as well as more mundane elements of the opportunity structure affording contact). Traditional social network analysis often arrived at these constructs either post hoc or tautologically: the pattern reflects an institution that gave it shape. But today, a variety of other means are available for discerning which mechanisms drive associations to take the form they take. A number of theories, which we cover in greater detail in coming chapters, posit some combination of exogenous and endogenous social forces that give rise to social structures and move them in predictable directions.

We divide these causal research questions in social network analysis into four broad types that vary in terms of what they seek to explain and how they approach social structure as part of that explanation. These types are summarized in Table 2.1.

The main theoretical division in views of social structure is between *connectionist* and *positional* approaches. As the names suggest, connectionist approaches see networks primarily through the lens of how nodes are tied through flows of important resources, objects, ideas, emotions, and so on. Positional approaches, in contrast, see structure through the lens of the ways specific parts of the structure tend to go together and interact through established roles. We do not see these as incommensurate, but instead as approaching social structures at somewhat different levels of abstraction in Hinde's view. Whereas connectionist views attend more to the lower levels of abstraction in interactions, relations, and groups, positional researchers seek to probe the higher-order abstractions of schemas and institutions that organize across many different interactions, relations, and groups. The former view is more bottom-up and locally emergent, and the latter is more top-down and globally prescriptive in its formulations.

To better illustrate the difference between connectionist and positional approaches, we offer a metaphor in which social structure is like the structure

TABLE 2.1 *Types of causal social-structural questions and social network research agendas*

Perspective	Networks as cause	Networks as consequence
Connectionist *Networks as pipes*	Diffusion Peer influence Social capital	Social integration Peer selection Segregation
Positional *Networks as roles*	Popularity effects Role behavior Network constraint	Exchange pattern Network stability Career paths

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of a watershed, a formation of land that separates flows of water into different rivers, basins, and seas. For connectionists, the key issue is understanding the flows as causes and consequences of important outcomes due to the transfer of important resources: the thriving of certain animal populations over others, the impact of climate change on these flows, and so on. For positional researchers, the key issue is the different parts of the watershed – estuaries, tributaries, floodplains, ridgelines, and so on – that come together and interact as specific roles or positions in the larger structure. Positional researchers therefore seek to understand the organizing principles and forces – such as erosion, accretion, and avulsion – that structure these positions in the process of creating these landscapes in generic ways and thus to understand why one watershed forms in a manner similar to or different from others.

As we will see throughout the chapters to follow, connectionist and positional approaches represent different but complementary endeavors in understanding social structure. These approaches become research projects when they seek to explain specific outcomes. For this reason, we further divide research into two typical approaches in terms of their dependent variable(s): (1) projects that seek to explain some important outcome using a structural explanation (networks as a cause), and (2) projects that seek to explain structure itself as the outcome of various structural and nonstructural explanations (networks as a consequence).

Let's walk through a few examples of these in somewhat more detail.

2.4.1 Networks as a Cause

What does it mean for a network to be a cause? Consider what is perhaps the core sociological concern: social inequality. How can social network analysis help explain an outcome like status attainment (i.e., having a more or less prestigious and well-paying occupation), which has largely been viewed through the lens of large-scale survey data? Social networks are clearly part of a causal explanation of inequality – but how?

One answer might be that social structure shapes who gets to interact with whom: individuals make connections with those around them and share important information and resources with one another in order to maintain power and authority in society. Connections matter. Karl Marx long ago argued that elites are more connected and better able to organize than are workers, who he saw as forming something more like a sack of potatoes, pushed up against one another by the capitalist system but not cohesive as a group "for itself." In contrast, elites, by sitting on the same corporate boards, attending the same country clubs, sending their kids to the same private schools, and so on, generate stronger ties that are conduits for important information and influence.

Personal networks are therefore a key aspect of *social capital* – that is, important new contacts or the information and resources contained within their direct and indirect contacts (Lin 2002). Access to social capital differs by race, class, and gender and has been shown to cause different rates of social mobility within and between occupational structures. The connectionist approach contends that a primary mechanism driving social life can be found in the contacts, interactions, and exchanges that directly connect one object to another, capturing important flows and influences across objects.

Referring back to Table 2.1, this top-left quadrant also includes, most notably, a rich research tradition on the diffusion of innovations, which we cover in Chapter 14. Here, the researcher is interested in how the structure of ties in a network facilitates or inhibits the transmission of ideas, information, and practices, leading to variable diffusion rates and diffusion patterns. A similarly large literature concerns the spread of infectious diseases through risk networks – for example, based on sexual contact or needle-sharing. In a world recently disrupted by the COVID-19 pandemic, such diffusion models have taken on an even greater import.

2.4.2 Networks as a Consequence

The top-right quadrant of Table 2.1 refers to connectionist approaches in which networks are outcomes of interest. The goal here is to explain the formation of network ties and larger structures, uncovering the factors leading to outcomes ranging from romantic bonds to corporate mergers, which we cover in Chapter 13. Mechanisms for such connections are frequently exogenous to the network structure and focus on not only physical proximity but also social proximity and *homophily*, the tendency for actors to interact with similar others in terms of shared cultural tastes, background characteristics, income level, and so on (McPherson, Smith-Lovin, & Cook 2001). Homophily is as close to a scientific law as sociology gets; although even homophily is highly probabilistic. A researcher may therefore ask how strong homophily is for important demographic dimensions, like age, gender, and race/ethnicity, or how homophily has changed over time (Smith, McPherson, & Smith-Lovin 2014). For example, do individuals tend to interact with people of the same race more than people of the same age?

Connectionist approaches also focus on endogenous factors, or those aspects of the network structure itself that led the structure to evolve in predictable ways. Here, for example, one might posit a "rule" in which "a friend of a friend is a friend," so that friendship ties elicit a type of transitive property that leads to the elaboration and thickening of the network structure. Bearman, Moody, and Stovel (2004) discovered a decidedly less intuitive "rule" of tie formation in the structure of the romantic network shown in

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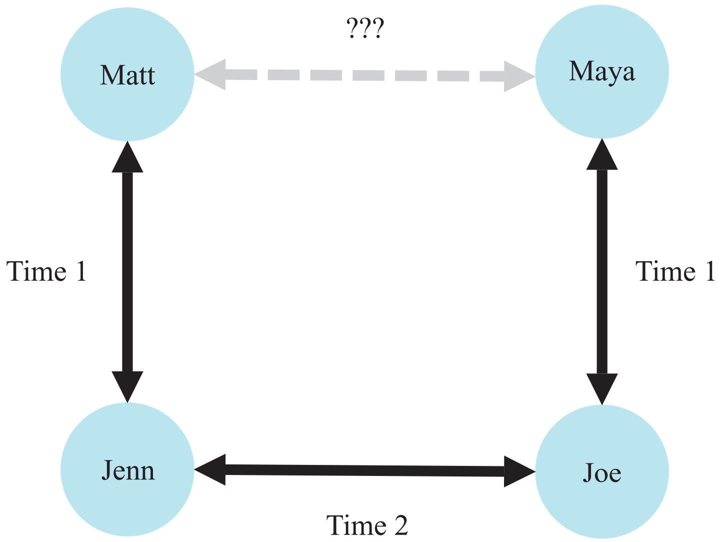

FIGURE 2.5 Hypothetical romantic network

Figure 1.2 – namely, its long chains. These chains appear to derive from an endogenous structural effect that largely lies outside of individuals' awareness: a tacit rule connected with the unpleasant feelings associated with dating the ex-partner of one's own ex-partner's current partner. Why? The discovery of this nonrandom element in the larger network structure is what inductively suggested an underlying rule of partnering, which the researchers dubbed the rule of "romantic leftovers": dating the ex-partner of your ex-partner's current partner is a largely forbidden structure. Individuals certainly do not articulate this as a rule. Nonetheless, the awkwardness of dating the ex of your ex's current partner clearly prevents such ties from forming in many scenarios (Figure 2.5).

The basic logic for this sort of explanation is that actors use the information embedded in their local ties to inform their likelihood of forming (or dropping) new (or old) ties. The set of features that come to play is up to the theorist but can include cognitive, emotional, and resource-based features.

2.4.3 Roles as Causes

Individuals fill multiple roles, generally defined by the institutions to which they belong. The family assigns kinship roles, educational institutions provide new roles, and work life provides yet another set of roles through which to relate. As depicted in Figure 2.6, the traditional family roles of parents, siblings, spouses, and so on are intuitive for those raised in a Western cultural context, although even here it is crucial to differentiate between the actual set of relations (e.g., romantic love, providing food for) from the named roles familiar to people (e.g., parent). A biological parent

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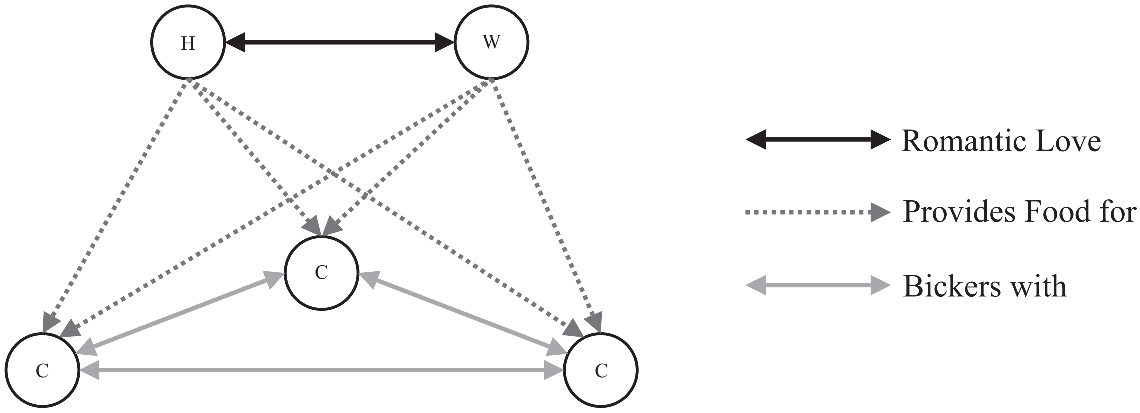

FIGURE 2.6 Role structure of a Western family unit

could, but need not, play the role of parent in the household as someone who provides for the child. As people shift from role to role, they become different people; they have "exits and entrances," as Shakespeare would say, and the parts they play are part of a continued biography as they move through social structure.

Moving to the bottom left quadrant in Table 2.1, the key question from this type of network approach is, How do structural roles in a network affect the thoughts, feelings, behaviors of specific individuals or groups? Seeing structural roles in relational terms is consistent with a long line of social psychological research using field experiments to introduce small differences into groups and finding that these differences crystallize into distinct role structures. Zimbardo's prison experiments, for example, demonstrated that the random assignment of students into the roles of "guards" and "prisoners" quickly induced thoughts, feelings, and behaviors in participants that were consistent with those experienced in a real prison. Rather than reflecting deeply held dispositions or personality traits, behavior may be shaped by more situational factors provided through roles.

The *positional* approach argues that the primary mechanisms driving social life can be found in these underlying social roles and in the norms attached to them. Together, these roles and norms guide patterns of interaction such that participants assume distinct social positions and coordinate patterns of association with occupants of other positions. An adult may be in a setting speaking to adolescents using a particular pattern of communication to broadcast information, and the youth may listen and receive it. The youth may perform work and submit it to the adult in exchange for evaluation, and so on. What they enact are teacher–student roles, and these place the adult and youth in distinct positions with particular forms of exchange, rights, and obligations. A positional approach examines the patterns of association and identifies underlying identities and identity boundaries (role relations) that evoke them. One can readily see this in other settings. As we described earlier, the game of baseball consists of a set of interactional performances based on the positions

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on the field and the expectations concerning how these are coordinated. Indeed, the early social psychologist George Herbert Mead (1934) thought of society as akin to the countless games people have internalized through their interactions and socialization.

How do such roles shape thoughts, feelings, or behaviors? Here, the answer is not so much about the flows of ideas, influence, and so on, but rather the expectations or norms that govern the behavior of individuals occupying specific positions. For example, being assigned to a leadership role comes with expectations and duties, which may give the role occupant more status, greater rewards, higher expectations of performance, and a greater belief in their competence. These may become internalized by the occupant and, in a kind of self-fulfilling prophecy, lead to their behaviors confirming expectations – or, following Lord Acton's dictum, the ineluctable corruption of a position with ultimate power. In an organizational structure, the personality traits often attributed to being a leader may have as much to do with having occupied leadership roles as they do with any inherent psychological differences. Similarly, the behaviors of individuals in similar positions may be expected to converge, even if those individuals are not directly socially connected. Here, individuals copy their peers' behaviors (e.g., dress, manner of speech, and adoption of new products), and the key mechanism is mimicry (of similarly positioned others) rather than direct influence and diffusion (Burt 1987).

2.4.4 Roles as Consequences

Finally, network scholars are interested in how specific roles and interconnected role structures emerge and change over time. Role structures may be intuitive to those who enact them, or social actors may find it difficult to know how their position fits within that larger structure, as Bearman (1997) showed in the case of bride exchanges among a set of Australian Aboriginal clans. Similarly, anthropologists have long puzzled over the origins of the Kula Ring in Melanesia, in which eighteen island communities exchange red and white shell adornments in two oppositely directed circles of which the individual communities are largely unaware (see Figure 2.7). Larger role structures may therefore emerge when local actors engage in local roles. Such forms of "generalized exchange" emerge from each island community's norms of gift exchange – which are based on notions of honor that help reproduce local social cohesion – combined with the island structure of adjacent communities with friendly relations and the exchanging of goods (other than the ceremonial adornments) through bartering.

Key questions for researchers operating in the bottom right quadrant of Table 2.1 concern examining roles as outcomes. What factors outside the observed network induce participants to construct and/or enact various role

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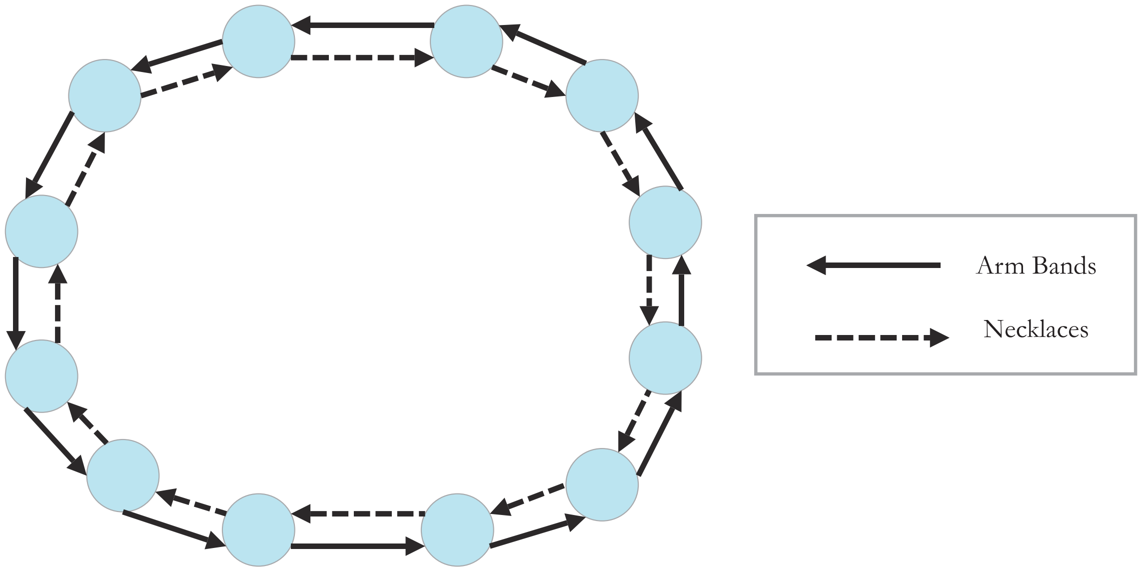

FIGURE 2.7 Role structure of the Trobriand Island Kula Ring exchange network

relations and assume various social positions? And how are these roles and positions sustained or altered over time?

Answering these questions presents unique challenges for the social network researcher. Without resorting to a functionalist tautology or wildly speculative evolutionary scenarios, in which the structure is seen to serve some purpose and to have evolved for some reason (e.g., social cohesion), network scholars often see structures as emerging from local patterns of interaction that become institutionalized and reified in cultural schemas and meanings. But testing such models can be daunting. It is difficult to perform the kind of field experiments employed by early social psychologists with similar questions. For example, in the Robbers Cave experiment, Sherif randomly assigned children in a summer camp to two different groups, which spontaneously developed their own identities (the "Eagles" and "Rattlers") and quickly became hostile toward each other (see Sherif et al. 1961). While Sherif demonstrated how quickly local social roles and intergroup conflict can emerge, finding data that can trace the *longue durée* of the emergence of larger societal roles is no small task. Bearman's (1993) study of the transformation of elite social structure over 300 years in Norfolk, England, represents a remarkable attempt to use historical network data to map the exogenous and endogenous factors that helped shift roles away from those rooted in local kinship toward those rooted in a national patron–client role structure, which was also expressed in an embrace of more abstract and universal ideologies for organizing more broadly.

Given the difficulties in gathering suitable data for examining the emergence and transformation of role structures, researchers have increasingly turned to agent-based simulations. As we will discuss in greater detail in subsequent chapters, various theories rooted in social psychology posit the emergence of role structures as routine outcomes of interaction and group dynamics. If these

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theories are correct, then small, even random, initial differences in attitudes among individuals – say, a preference for nine-pin bowling over ten-pin bowling – can generate a Robbers Cave–like scenario of two oppositional subgroups (the “nines” vs. the “tens”). Other models suggest that even more fragmented role structures or hierarchies of subgroups, like those frequently seen in high schools based on popularity, are emergent role structures rooted in self-organizing principles of individual agents and their interactions (Goldberg & Stein 2018; Macy et al. 2003; Mäs & Flache 2013; Szell, Lambiotte, & Thurner 2010). Although provocative, these models await more rigorous empirical testing to probe their merits.

2.5 SOME CAVEATS

Our goal in this chapter has been to orient the reader to a serviceable theory of social structure that can help guide practical research efforts in social network analysis. For more detailed accounts of social structure, we refer the reader to a number of theoretical approaches that have informed social network research and our overview (e.g., Blau 1977; Burt 1982, 2005; Crossley 2010; Emirbayer & Mische 1998; Fuhse 2009; Martin 2009a, 2011; Sewell 1992; White 1992). Throughout this volume, we will return to the layered description of social structure, and we will fill in some more details as we go. Here, we conclude this chapter by further noting two main caveats in rendering social structure into knowledge.

2.5.1 An Evolving Heuristic

First, it is important to emphasize that the four quadrants in Table 2.1 should be taken as a heuristic for orienting network research, not as a tidy summary of any individual project or scholar’s research agenda. The connectionist and positional approaches are commensurable and often work in tandem to offer a fuller understanding of structural phenomena (Nadel 1957). Many research projects are not isolated within one quadrant, but instead seek connections among them. For example, researchers have drawn greater attention to the endogenous mechanisms from role structures themselves that are self-sustaining, connecting role-based approaches in the two lower quadrants. In this light, being one of the popular students in a high school has a “sticky” quality: they are treated differently, often in ways that enhance their visibility and a belief in their specialness, which tends to reinforce the high school pecking order over time. Likewise, a reciprocal causality exists across the top two quadrants, which see networks primarily as pipes for important resources. Clearly, one’s social capital can lead to opportunities to form new ties, which in turn lead to even greater social capital. Many other possible projects could link

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the quadrants, and as we discuss in the concluding chapter of this book (Chapter 16), this table could be expanded and altered in various ways as social network analysis moves into its next phase.

2.5.2 Institutional Reproduction and Change

A second caveat concerns how social structure appears to overwhelm human agency. Any view of social structure necessarily emphasizes institutions and the reproduction of largely stable patterns of activity. What is less visible is how social structures often contain considerable ambiguity and complexity that render them less deterministic than one might think (Merton 1976).

For theorists like Sewell (1992) and Giddens (1986), social structures tend to form by a process of *structuration*, or a combination of *structure* and *agency*, which we cover in greater detail in Chapter 6. For Sewell, this process is described as the co-evolution of schemas (like roles or identities) and manifest relations and interactions (which are used as resources to cue schemas). Interestingly, though, Sewell does not see this process as a static reproduction. Social interaction patterns reproduce themselves, but they also change and drift. Building on Sewell's insights from a social network perspective, we see two main aspects of social structure as affording a nondeterministic view of social structure: (1) multilevel (or multimodal) networks and (2) multilayered (or multiplex) networks.

2.5.2.1 Multilevel or Multimodal Networks

As we show in Figure 2.3, social structure exists at various levels, from events to organizations. Networks also exist through affiliations, such as group memberships, that are depicted as multimodal networks. We can (and will) extend such levels to include various roles and even categories that organize human thought and action. The embeddedness of social structure allows for a certain nondeterminism because various levels of social structure are only approximately nested. For example, in our view of the high school, levels of structure are largely hierarchical; however, many informal interactions and relations are likely to exist that defy an easy determinism. For example, different social-structural positions in a school can draw on the same resource for cross-purposes. Popular students and teachers both want to assume central roles in classrooms and are often at cross-purposes. As such, in some classes, the teacher may coopt popular students by giving them central roles in lessons in exchange for their compliance and a degree of latitude. Social structure can therefore encourage adaptive alliances rather than simply reproducing the status quo. In short, different levels of social structure may have their own cross-cutting pressures that render multiple social actions equally feasible, rather than determining a single and clear-cut role behavior.

2.5.2.2 *Multilayered or Multiplex Networks*

Social structure emerges from many different types of ties. The multilayeredness or multiplexity of ties affords nondeterministic action. Sometimes the same role relations can be used as a resource in other situations: friendship, for example, can be used for completing schoolwork and satisfying student role demands as well as for developing dating relations. As such, the same resources can have multiple meanings and uses, and audiences may read and interpret the same resource differently. To know a rule or role is to be able to transpose and extend its use to new situations (other classrooms), but matching existing schemas to new situations can pose indeterminacies that require creativity and social learning, intersubjective agreements, and coordination. Teachers must adapt to teach students who are unlike the ones they have taught before. Thus, individuals have unique biographies (distinct histories and locations) within a multilayered network landscape, and this structural complexity leads to opportunities for creative action.

2.6 SUMMARY

This chapter has oriented the reader's imagination to an expansive view of social structure, from interactions to institutions. Based on this structural account, we have proposed a broad research agenda for how to coordinate efforts of social network analysis in descriptive and predictive accounts. Social structure is far too complex (at least for our brains) to see in its entirety, let alone to know where social structure comes from and what it may be doing. Social network analysis is a powerful approach in bringing together these efforts and drawing on related theories. We will return to our view of social structure (Figure 2.3), especially in Part II on "Seeing Structure," and the causal research agendas (Table 2.1), especially in Part III on "Making Structural Predictions." In many of the following chapters, we show how social network analysis has filled in this picture and where work still remains to be done.

SUGGESTED FURTHER READING

General Theoretical Formulations. The theoretical foundations of social network analysis are deep, and we cannot hope to cover all pieces here. Rather, we want to highlight a handful of key pieces that have helped shape how we view the field.

Blau, Peter M. 1977. *Inequality and Heterogeneity: A Primitive Theory of Social Structure*. New York: Free Press. (Provides the foundation for work on homophily and opportunity structure; linking social relations, interests, and the intersection social characteristics. See also CATNETs by Harrison White.)

Bott, Elizabeth. 1957. *Family and Social Network*. New York: Taylor & Francis. (A lovely examination of the effects of gender and network segregation on role performance. Foundational work for the study of family and normative pressures in networks.)

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Castells, Manuel. 2009. *The Rise of Network Society*. Hoboken, NJ: Wiley-Blackwell. (Argues that modern social life is fundamentally changed by the advent of "information and communication technologies," which allows for the widespread globalization of networks. See also the rich literature on World Systems Theory, particularly the empirical work of Christopher Chase-Dunn and colleagues.)

Coleman, James S. 1994. *The Foundations of Social Theory*. Cambridge, MA: Belknap of Harvard University Press. (Attempts to build a social theory based on purposive action constrained by relational opportunities. Exemplar case of mathematical theorizing.)

Emirbayer, Mustafa. 1997. "Manifesto for a Relational Sociology." *American Journal of Sociology* 103: 281–317. (Argues that the essence of any valid model for social structure has to be relational; provides a deep foundation for network models of social life.)

Hinde, Robert A. 1976. "Interactions, Relationships and Social Structure." *Man* 11(1): 1–17. (Describes primate social structures from concrete interactions up to specific relations and types of relations and roles, groups, and types of groups. Basic framework of abstraction/concreteness and units of analysis to social structure.)

Martin, John Levi. 2009. *Social Structures*. Princeton, NJ: Princeton University Press. (Provides a theoretical explanation of how types of relations constrain/construct types of aggregate social structures.)

McPherson, J. Miller. 1983. "An Ecology of Affiliation." *American Sociological Review* 48(4): 519–32. (A key work summarizing the macrostructural homophily perspective on social organization. See also McPherson and Ranger-Moore 1991 and anything by Bruce Mayhew.)

McPherson, J. Miller, and James Ranger-Moore. 1991. "Evolution on a Dancing Landscape: Organizations and Networks in Dynamic Blau Space." *Social Forces* 70: 19–42. (See the note to McPherson 1983.)

Nadel, Siegfried Frederick. 1957. *The Theory of Social Structure*. Glencoe, IL: Free Press. (Presents roles as interrelated systems of social expectations and obligations.)

Sewell, William H. 1992. "A Theory of Structure: Duality, Agency, and Transformation." *American Journal of Sociology* 98: 1–29. (Integrates agency as a type of structure, drawing on insights from Giddens and Bourdieu.)

Simmel, Georg. 1950. *The Sociology of Georg Simmel*. Glencoe, IL: Free Press. (Overview of many elemental ideas of social networks concerning relations, dyads, triads, and organizations, as well as issues of size, reciprocity, and *tertius gaudens*.)

2010 [1908]. *Conflict and the Web of Group Affiliations*. Glencoe, IL: Free Press. (Describes how overlapping social circles can be theorized and studied and how they organize in society.)

Somers, Margaret R. 1994. "The Narrative Constitution of Identity: A Relational and Network Approach." *Theory and Society* 23: 605–49. (Offers a theoretically informed account of relational sociology and how identities are more akin to narratives and stories.)

Turner, Jonathan H. 1978. *The Structure of Sociological Theory*, 7th ed. Belmont, CA: Wadsworth. (A general theory reader with a rich discussion of the structural-functional roots of early role-behavior models.)

White, Harrison C. 1992. *Identity and Control: A Structural Theory of Social Action*. Princeton, NJ: Princeton University Press. (An effort to present a synthetic theory of social structure that draws on cultural turn sociology [interactionist and sociolinguistic] and social network theories concerning roles, but with extensions to macrostructures and processes, like disciplines and network domains.)

Illustrative Empirical Works. Social network analysis is characterized by a tight linkage between theory and method, as researchers develop tools to solve empirical questions. As such, some of the best general network theory has been developed in the context of empirical investigations.

Bainbridge, William Sims. 2020. *The Social Structure of Online Communities*. New York: Cambridge University Press. (For obvious reasons, novices often conflate social media and social networks; this work provides a clear example of how to apply deep network thinking to online communities without falling prey to computational approaches that speak only to scale and not substance.)

Bearman, Peter. 1997. "Generalized Exchange." *American Journal of Sociology* 102: 1383–415. (Using blockmodels of kinship exchange, this paper provides a clear example of how people enact social structure, even if they cannot articulate why they do so.)

Gibson, David R. 2003. "Participation Shifts: Order and Differentiation in Group Conversation." *Social Forces* 81: 1335–80. (Empirical example of how relational events of different sequential turn-taking strategies inform the development of roles and networks.)

Gould, Roger V. 1991. "Multiple Networks and Mobilization in the Paris Commune, 1871." *American Sociological Review* 56(6): 716–29. (Uses data on deaths and fighting across neighborhoods and blockmodel analysis to reveal the underlying structural basis of conflict.)

Mizruchi, Mark S., and Michael Schwartz (eds.). 1988. *Intercorporate Relations: The Structural Analysis of Business*. New York: Cambridge University Press. (A nice early edited volume on network approaches to economic action.)

Padgett, John F., and Christopher K. Ansell. 1993. "Robust Action and the Rise of the Medici, 1400–1434." *American Journal of Sociology* 98(6): 1259–319. (A historical illustration of how a political system emerges out of the interactions and alliance systems of Renaissance elite family politics.)

3

What Is a Social Network?

We outline the core concepts of network analysis, issues of research design, and data collection for different structural questions.

3.1 THE NETWORK IDEA

Poets and sages have long spoken of the interconnectedness of things. Although most had structural imaginations, they were not network scholars because they lacked the institutions and formalization necessary to channel this imagination into an empirical research agenda they could share with others. In the absence of scientific research tools, their imaginations were expressed through metaphor alone, and their ideas could not be built into a collective endeavor of discovery and refinement. To move from simply thinking structurally to actually seeing and formalizing structural analysis, one needs a common language and set of symbols to communicate with others working on similar problems and to transmit these findings to others over successive generations of scholars. Network analysis is one way researchers have operationalized thinking structurally.

Moving from structural imaginations to network analysis begins with visual and mathematical abstractions of relational phenomena that are otherwise difficult, if not impossible, to capture verbally. Consider the following historical example, a conundrum that remained unresolved until the advent of a network-like approach in 1736. For centuries, the city of Königsberg in Prussia had seven bridges connecting the mainland and two islands, as shown in Figure 3.1. This unique geography led people with too much free time to ask whether it was possible to traverse all seven bridges *only one time each* and yet touch every land mass in the city. With thousands of possible "walks," this debate kept people busy for a while, until mathematician Leonhard Euler resolved the issue by devising a network abstraction that removed the content of the

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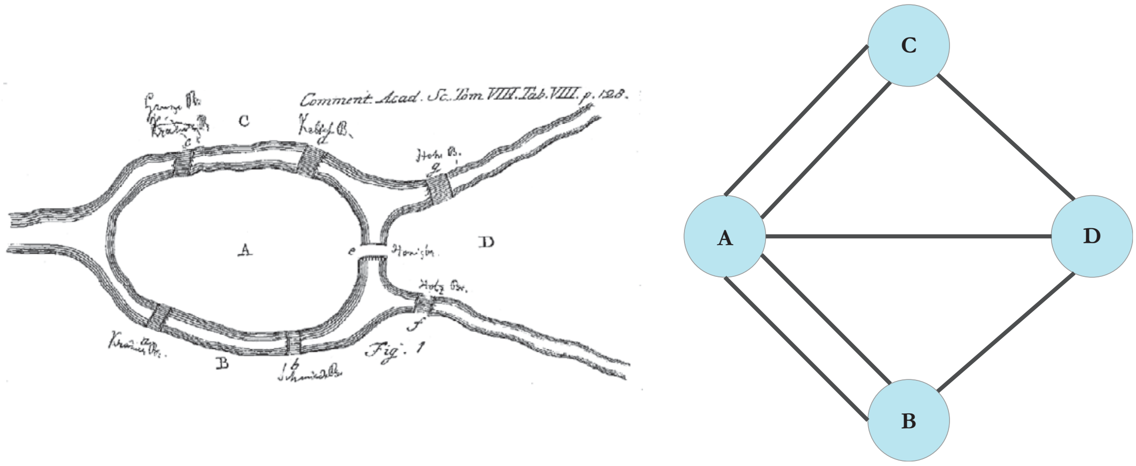

FIGURE 3.1 The Königsberg bridge problem and its graph representation

puzzle (bridges and landmasses), leaving only its formal properties (elements and linkages).

In the second image, the landmasses become elements (or objects) represented as points A through D, and the bridges become lines (or links). Once one is no longer puzzling about all the ways one could walk over all the bridges and instead examines the formal aspects of a structure, it becomes easier to see and resolve the problem. In fact, Euler demonstrated that no solution exists, given the overall network structure: because each point has an odd number of lines connecting it to other points, all the complete walks eventually lead to being trapped at a point after having already traversed all the other paths leading to or from that point. Euler's solution helped initiate a new branch of mathematics: graph theory. Network researchers would later use and develop graph theory to understand more substantively important and complex structures. As with the preceding example, the key network idea rests on representing a structure as an interconnected set of points. Doing so allows the researcher to draw an image in an intuitive way and to use that visualization to communicate with others.

As Euler demonstrated, network analysis starts with concrete interactions (here, the rather stable "interaction" of bridges connecting land masses) and offers an abstracted picture of objects and links within a more or less integrated whole – a structure. In this example, one might see an implicit connectionism at the heart of network analysis: people move across the stable structure of the interconnected system of bridges. Such a tendency for focusing on things flowing through ties as the basis of the network persists today. Most social network research focuses on interactions and relations, which are commonly assumed to be built from important exchanges. But as we will see, moving to greater levels of abstraction reveals higher-order structures that may not be well represented as flows.

A shared set of abstractions is at the heart of any science. The abstractions developed in the study of structures help make it possible to analyze systems in tractable ways. The generalizable quality of these abstractions also means that the same or similar methods can be used for very different substantive problems, connecting scholars across fields with shared structural interests but different theoretical and substantive applications. Of course, this does not mean that network researchers can rely solely on methods and data; nor does it mean that an explanation that works in one setting (adolescents in a high school) will necessarily work for another (neural networks in a brain). Across all fields, one must apply the relevant theories and background knowledge to the particular network(s) at hand. This is why we introduced social-structural theory in the opening chapters: a network is an abstraction that requires substantive and theoretical understanding to be rendered into useful information.

With this caveat in mind, we proceed with a more detailed discussion of the core aspects of network analysis while offering some substantive examples from sociological studies. In Chapter 4, we will revisit many conceptual issues and give more insight into the theoretical and practical trade-offs involved in the operationalization and collection of network data.

3.2 CONCEPTUALIZING NODES AND TIES

What are the objects and links forming a network structure? Network researchers use the term *nodes* to refer to the objects in a structure but sometimes refer to these using the graph-theoretic term *vertices*. Nodes are generally represented as points in a figure. The relationships among the nodes are most frequently called *ties*, although the more mathematical terms *edges* or *arcs* are also used. For example, in Figure 3.2, the nodes are represented as circles labeled A–F, and the edges are presented as lines between connected nodes. Network scholars usually refer to such visualizations as *graphs*. Let's assume that these edges and nodes in a graph represent friendship ties between children at camp. We see that A is friends with B, E, and D; B is friends with A and E; and so on. When two nodes are connected, they are called *adjacent*. Thus, a network can be formally defined as two linked *sets* (i.e., collections of things): one of nodes and one of ties.[1] How one conceptualizes the nodes and ties has consequences for every other aspect of the research that follows. At a low level of abstraction, ties might be momentary interactions captured in small slices of time. These interactions might be aggregated into higher levels

[1] Technically, the edge set (a) must be a subset of all pairs of nodes and (b) could be empty. Practically, though, a network without ties would rarely be interesting.

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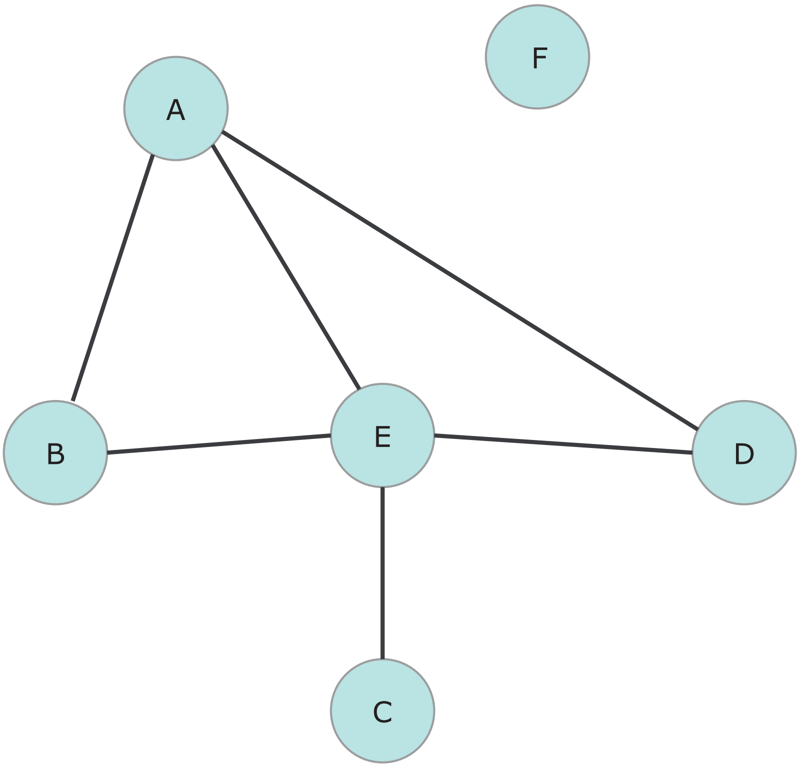

FIGURE 3.2 A basic network structure

to infer relations. For example, the network might represent an aggregate of playful interactions during recess over the period of a week, which would in most cases be a reasonable operationalization of "playmate" or even "friend."

In most cases, identifying the nodes in a structure is the more straightforward task because the nodes are more stable elements, whereas ties can be formed or altered. In social network analysis, the nodes are social actors at various levels of aggregation, from individuals to groups, organizations, or even nations. However, determining *where* to define the boundaries around the *node set* – the collection of nodes forming the network – is often a difficult task and may even involve an understanding of *when* the nodes exist such that they can be considered part of the same structure. Once again, this often entails a substantive and theoretical understanding of the empirical domain of interest. A researcher using ethnographic methods to study peer influence in a high school may know that a group of adolescents regularly meets after school on the playground. Thus, this may be a suitable space and time for determining the nodes in a network, as these individuals interact and perhaps perceive themselves as a group. Obviously, this will be an incomplete structure of all the peer influences for these individuals, but as we discuss in greater detail when outlining data collection strategies, nearly all network studies are partial slices of much larger and more complex structures.

Defining the ties among nodes can be tricky. In social network analysis, ties most frequently represent *positive* social relations in which two nodes are tied through interactions and role relations that entail positive feelings, intimacy, and supportive interactions. These relations may be observed by the researcher and/or perceived by the subjects. Thus, social networks are often ones of kinship, friendship, and exchange. That said, there are *negative* social relations that concern negative attitudes, directed feelings such as disliking, and conflictual interactions such as bullying. Typically, people gravitate toward positive ties and avoid negative ones. Negative social ties are traditionally understudied and are an active area of current research (see Kitts 2014).

From a connectionist standpoint, ties often represent relations built from flows and exchanges of real things in the world that might move between individuals (e.g., money, diseases) or between higher-level social actors (e.g., people moving between jobs or countries). Flows may also be more abstract things, such as the movement of ideas, beliefs, or even emotions. From a more positional account of structure, ties may have less to do with actual flows through connections than they do with occupying a similar location in a structure (based on rights and responsibilities), and this may give rise to similar pressures or awareness of others (Burt 1987). Social ties can be roughly considered as falling into four types (Borgatti et al. 2009):

- *Memberships,* in which nodes are located in the same regions in physical and social space (e.g., same neighborhoods, same department, or same club).
- *Relations,* in which nodes operate within a system of roles (e.g., father of, friend of, or teacher of) and have cognitive or affective orientations toward one another.
- *Interactions,* in which concrete interactions occur between nodes (e.g., advice, romance, or bullying).
- *Flows,* in which nodes transfer some material or cultural object, goods, information, or influence (e.g., ideas, beliefs, or practices).

Importantly, a network study can include more than one type of tie. For example, the researcher may have information on similarities, like shared clubs, as well as more affective relations, like friendship (e.g., people could be part of the same club but not be friends). This yields a network with multiple types of ties, or a *multiplex network*. Also important is that the ties can be dynamic: they may be made and unmade over time, often with some involvement of the nodes themselves (e.g., two people may choose not to remain friends). To build a more dynamic view, researchers often take many different snapshots of structures over a period divided into years, months, weeks, or even minutes. With streaming data from visual or online sources, it is also possible to gauge networks in continuous time. Having such longitudinal network data offers considerable advantages over a single cross section. Gathering a more realistic view of the underlying reality of structure as dynamic and layered is the goal of most research, but of course no single project can accomplish this completely.

Each individual who is connected within a network has a number of ties, which is referred to as their *degree*. For example, in Figure 3.2, individual E has a degree of four; individual A has a degree of three; individuals B and D degrees of two; individual C a degree of one; and individual F a degree of zero. As we will discuss more in the next section of this chapter, networks can also be *directed* so that ties may not be symmetric. When network ties are asymmetric, the number of ties directed at an individual is known as their *in-degree*, and the number of ties directed outward is their *out-degree*.

Network ties can also vary in their overall *strength*. People have acquaintances, friends, and close friends; corporate donations to political campaigns can vary in amount; and countries exchange goods representing different amounts of their GDP. Tie strength can also have a negative/positive valence: for example, hatred, dislike, neutral, like, and love. Social connections are usually thought of as positive relationships, but the social world also contains antagonistic ties, and recognizing them is important for understanding the full functioning of a social system. Such signed (i.e., positive and negative) data raise questions of group stability, balance, conflict, and the maintenance of identities, where group members have positive ties to fellow members and negative ties to outsiders. With so many conceptual issues at stake, the validity of ties and how they are measured are areas of particular concern. A valid network study will need to clearly justify the inherent trade-offs involved in the measurement of ties.

3.3 TYPES OF NETWORKS

Constructing a network of actors and ties, or nodes and edges, seems like the result of an objective set of decisions. However, keep in mind that one is already engaging in some theorizing when one defines the nodes and edges, determining why certain nodes are important for the structure and what holds the structure together. Sometimes the researcher is able to gather data that directly address the underlying processes creating the structure one is seeking to measure. At other times, one uses *proxies* with data that are less than ideal for gauging those underlying processes. All data have some error, and no data set will fully capture a social structure. As such, the first step in network analysis is to have some sense of the core aspects of the social structure one intends to study (which actors and types of ties). The next step is to make sure the type of data collected matches one's underlying research goals. In so doing, one acquires data most reflective of the structure in question, and the ensuing information and knowledge presented will offer important insights into social reality.

Network data, as sets of nodes and ties, come together to form structures of two main types: (1) direct ties in *one-mode* networks and (2) affiliations in

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two-mode networks.[2] For example, person *i* reports spending time with person *j*, or the researcher directly observes this. Such relations are called one-mode because they consist of only one type of node (people in this case) with no *a priori* restriction on ties between subclasses of nodes. Ties can also be created because nodes of one type share connections to nodes of a second type. For example, if person *i* belongs to the same country club as person *j*, then we can link them by a "shared country club" tie. In this case, the *ij* tie goes through a mutual association with a club. Such relations are called *two-mode* because they link nodes of one type (people in this case) through nodes of a second type (clubs in this case). These networks admit to a special "duality," such that either type of node can be connected through the other type. In our example, then, one could connect various country clubs to one another via their shared memberships just as easily as connecting people through their associations. As we will discuss in greater detail in Chapter 11, the simultaneous connections existing among two orders of nodes – such as people tied through country clubs and country clubs tied through people – is a common structural duality that motivates much network research. It is also possible, and increasingly common, to analyze networks consisting of a mix of one-mode and two-mode relations; for example, one may have interactions between students (one-mode), alliances between clubs (e.g., co-sponsoring events; one-mode), and students linked to clubs through membership (two-mode).

Networks can be either *directed* or *undirected*, depending on the nature of the network. Ties are directed when they indicate an orientation or flow from *i* to *j*, such as when *i* indicates that *j* is a friend or when a researcher observes something being transmitted from *i* to *j*. In the case of directed ties, it is possible for the tie to go from *i* to *j* but not necessarily from *j* to *i*. For example, person *i* may loan money to person *j*, but *j* may never have loaned *i* any money. More generally, one can define asymmetric ties as those in which there is a tie from *i* to *j* but not from *j* to *i*. A symmetric tie is simply the case in which an *ij* tie exists and a *ji* tie exists. Symmetric ties capture instances of reciprocity, such that the relation from *i* to *j* implies the same relation between *j* and *i*. For example, there may be expectations of reciprocity for gift giving: if you give a gift to me, I may feel obligated to give a gift to you. But this is an empirical question.

The network type thus depends on the logically possible distribution of tie types. In an *undirected* network, all the ties are assumed to be symmetric ($ij = ji$). A network based on consensual online friendships on Facebook, for example, is undirected; if *i* accepts *j*'s friendship request, then both appear in

[2] Multilevel or multimodal networks are also possible (e.g., Stofer 2008). Because these represent more of a frontier in network science, we discuss these in our concluding chapter (Chapter 16).

each other's "friends" list. A network based on joint activity, such as "neighbors" is also undirected (i.e., if *i* is *j*'s neighbor then *j* must be *i*'s neighbor). Alternatively, if the ties must be asymmetric, then the graph is *antisymmetric*. The most common antisymmetric networks in social science reflect clear constraints in the setting, such as formal hierarchical orderings ("command" in the military), or historical constraints, such as citations where it is impossible for a past paper to cite a future paper because the future paper does not exist at the time the first paper is written. A special case of an antisymmetric network is the directed acyclic graph (DAG). A DAG is an antisymmetric network with no directed cycles. The lack of cycles implies a (partial) ordering of nodes based on their position in the sequence of relations (e.g., General ➔ Colonel ➔ Major ➔ Captain), which has proved important for many social science applications, particularly in narrative networks (e.g., Bearman, Faris, & Moody 1999; Bearman & Stovel 2000) and causal modeling (e.g., Textor et al. 2016). If the setting allows for either symmetric or asymmetric relations, the network is "directed," and the extent of reciprocity is often an empirical question of interest.

Combining directionality with notions of which ties are valued or unvalued (capturing the strength or valence of the tie) creates a typology of four networks, which can be visualized as shown in Figure 3.3. Networks where ties either do or do not exist are considered *binary* versus ties that have an inherent strength, which are *valued*.

The directed networks have arrows indicating a tie from one node to another, whereas the undirected networks do not (i.e., all ties are symmetric). The valued networks extend the binary networks by placing weights or strengths on the ties (e.g., strength of friendship, amount of loan), represented

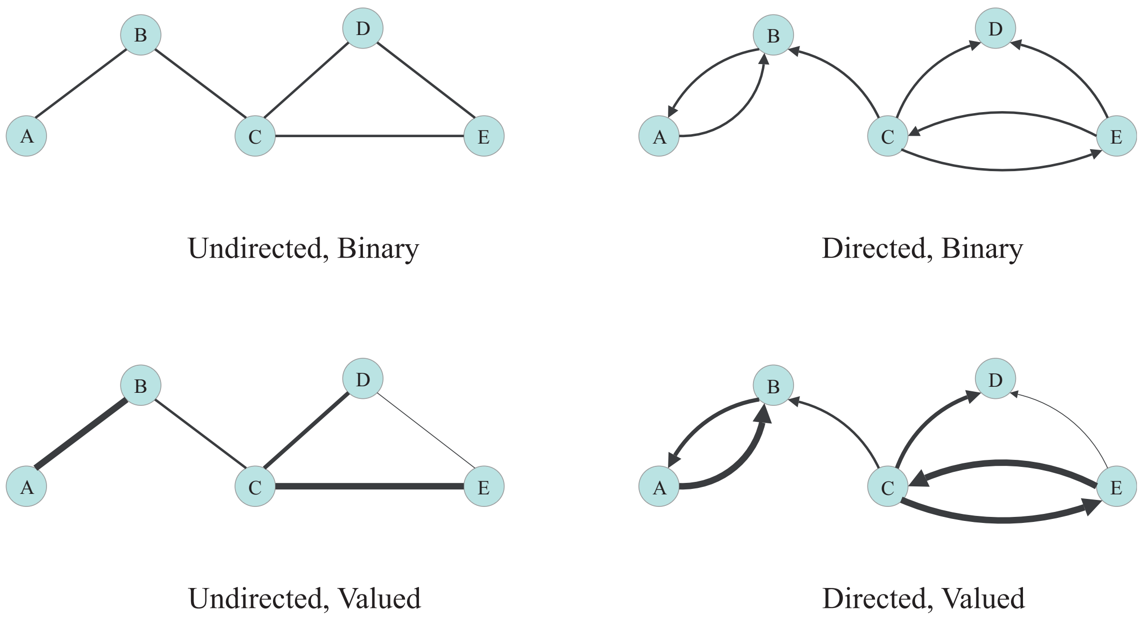

FIGURE 3.3 Four types of networks

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in Figure 3.3 by thinner or wider lines. One could also imagine an analogous picture that includes more than one type of tie (e.g., friendship and loan relations among the same nodes).

We can further illustrate a one-mode network with an example drawn from archival research on the relations among prominent families in Florence, Italy, during the Renaissance (Padgett & Ansell 1993). During this period in Italy, family and business were very closely aligned; thus, as represented in Figure 3.4,

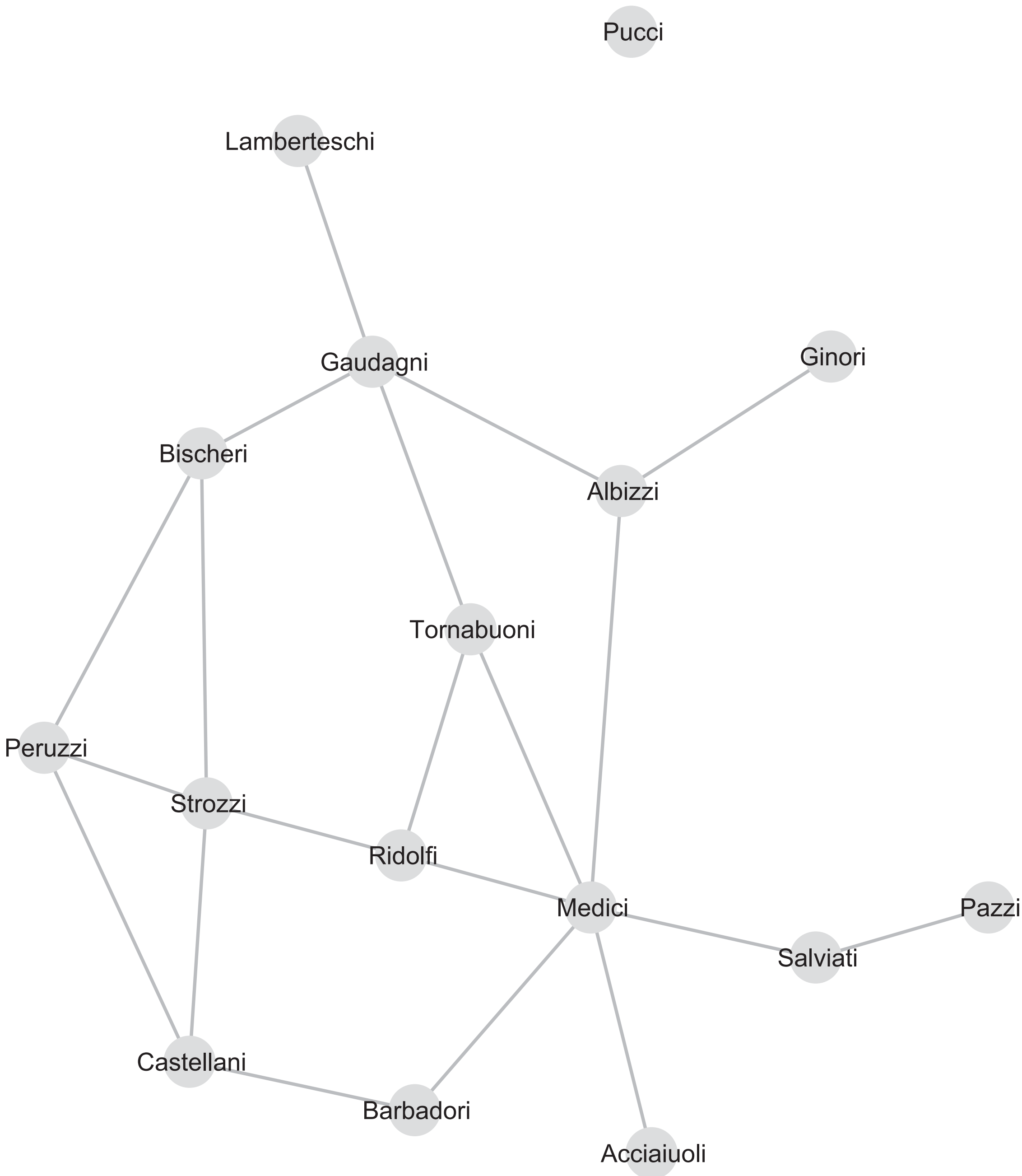

FIGURE 3.4 Kinship/business relations among Florentine families in the Renaissance

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the nodes are families, and the ties are relations based on intermarriages between families. The researchers chose marriages because these ties represent primary group affiliations, or strong ties, guiding other sorts of associations. Because marriage ties represent a mutual tie across families, they are symmetric. Here, the network image is binary, indicating that a tie is either present or absent; that is, the value of the tie (e.g., the number of intermarriages) is not taken into account. Taken as a whole, the set of families and their interdependencies can be considered a family network of municipal elites.

Most people will recognize the Medici name because of the family's prominent associations with the patronage of great works of Renaissance art. The network reveals the Medici's interesting and important position within the structure of Florentine families. Think of the families as landmasses and the kinship/business ties as bridges in the Euler example presented earlier. If one were to try to "walk" to each family, which ones would one likely have to traverse more than once to reach everyone else? The formal property of being between other entities is common and important to many structures. Substantive knowledge about what is likely "flowing" through these ties is also key. Because important resources, information, and the capacity to exert social influence depend on a family's structural location, an additional structural property to note here is the extent to which each family is dependent on certain families to access both other families and the information that flows between them. In this regard, the Medici are likely to be most powerful not only because they are between many families but also because many of those families are dependent on the Medici family to reach others.

Next, let's examine affiliation network data and their two-mode graph visualization. The network shown in Figure 3.5 is based on data collected by Davis in the 1930s (see Davis et al. 2009[1941]) as part of a community study in the southern United States. The figure shows eighteen women (red nodes) and the network of their having attended fourteen social events (green nodes). The women are not directly connected to one another, nor are the social events directly connected to one another. Instead, they are linked indirectly through the other type of node (i.e., events). Women are affiliated with one another by attending the same social events, and social events are connected by sharing one or more female attendees. As with the Florentine families, the southern women's network reveals something about the structure of a community, and the relational positions of actors that correspond with their social importance. We will formalize this intuition in much greater detail in Chapter 9, when we discuss the concept and measures of network centrality. Here, we note that visual inspection alone can be a powerful tool. We can roughly conclude that there appear to be two different groups of women: the "Evelyn" set on the left and the "Nora" set on the right. Looking at the events, we can see that Events 1–5 correspond most exclusively with the "Evelyn" set, and Events 10–14 correspond with the "Nora" set. Events 6–9, in the middle of the graph, are the most important in terms of integrating the community because they are the events that women from both groups attended.

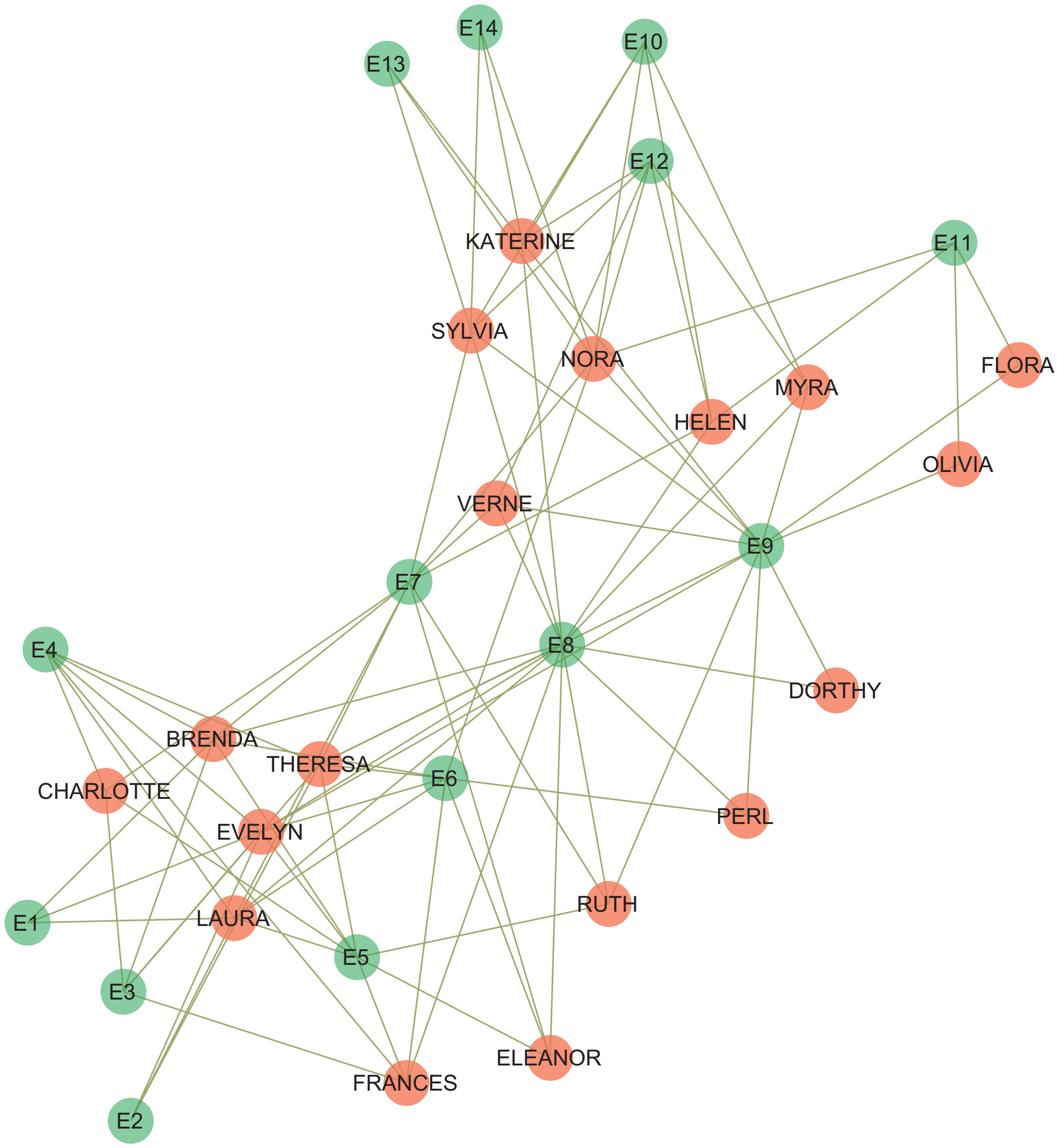

FIGURE 3.5 Southern women and their social event attendance

The decision to employ either a one-mode or two-mode network is highly dependent on the researcher's theoretical approach and empirical questions. For example, a researcher taking a connectionist-based view of networks, in which nodes transmit or represent flows of information to one another, will usually choose a one-mode network, such as the Florentine families. A one-mode network captures the direct connections between actors and thus can be used to infer the diffusion potential (or flows) across the network. In contrast, a two-mode network, in which individuals are indirectly linked through their affiliations, is less precise in gauging flows. For example, in Figure 3.5, without any indication of a direct tie between Laura and Verne, the researcher may

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incorrectly infer flow and contact between them, when the flow may actually be coming from their joint association with the 8th social event (E8). Affiliation networks are, however, very practical because of the relative ease of collection, and they sometimes represent the only option available to a researcher. For example, researchers often use two-mode networks to study intellectual or creative influences in project-based teams, such as when actors appear in the same film or film writers collaborate on the same script.

3.4 GETTING A FEEL FOR NETWORKS

Many network measures build on the notions of flow between actors, and it is useful to have an intuitive idea of these measures before trying to apply them in more complicated settings. Figure 3.6 presents a simple network of five people in which the tie of interest is "seeks advice from." For example, Sally frequently seeks advice from Harry and Tim frequently seeks advice from Sally, Harry, and Mary; and so on. We will use this toy network to demonstrate how network ideas – such as walks, paths, and distance – can be measured from the basic data describing the connections between each actor.

Walks capture the potential flows between actors when the item of interest can visit an actor more than once. A little more formally, walks capture any sequences of nodes connecting i to j, with no restrictions on the number of times the walk moves over a given actor. For example, Figure 3.7 shows walks of length 3 starting from Tim. Looking at the very first branch, one can see that Tim seeks advice from Sally, who seeks advice from Harry, who seeks advice from Sally. Overall, there are three different walks of length 3 connecting Harry to Sally: (1) Tim → Sally → Harry → Sally; (2) Tim → Sally → Tim → Sally; and (3) Tim → Harry → Tim → Sally. Thus, information (e.g., rumors) discussed by Tim could reach Sally in three different ways. One must follow the direction of the tie in calculating the length of the walk.

Paths are similar to walks but add the restriction that the sequence of nodes can go over each actor only once. This restriction would be appropriate if one thought that little would be added by hearing the information from the same person again. Substantively, walks capture something akin to echo chambers, whereas paths are more like the game of telephone. In this case, there is one path of length 1 and one path of length 2 connecting Tim to Sally (Tim → Sally; Tim → Harry → Sally). Note that all paths are walks, but not all walks are paths.

Finally, distance is generally calculated as the shortest path connecting actor i to actor j. The length of the shortest distance between two actors in a network is called a *geodesic*.[3] In the case of Tim to Sally, distance is 1 because a path of length

[3] Other ways of measuring distance are possible, such as the expected number of ties a random walker would have to traverse to connect two nodes, but these are less commonly used in social network analysis.

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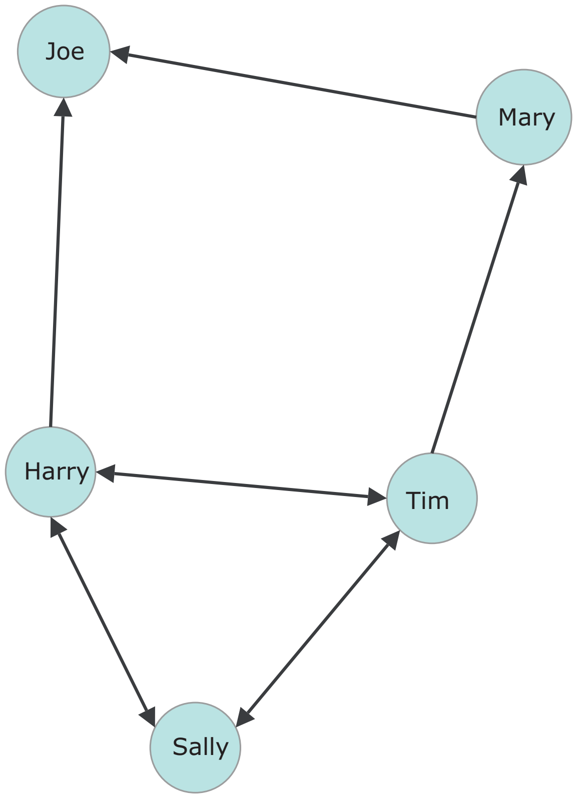

FIGURE 3.6 Example graph and basic network definitions

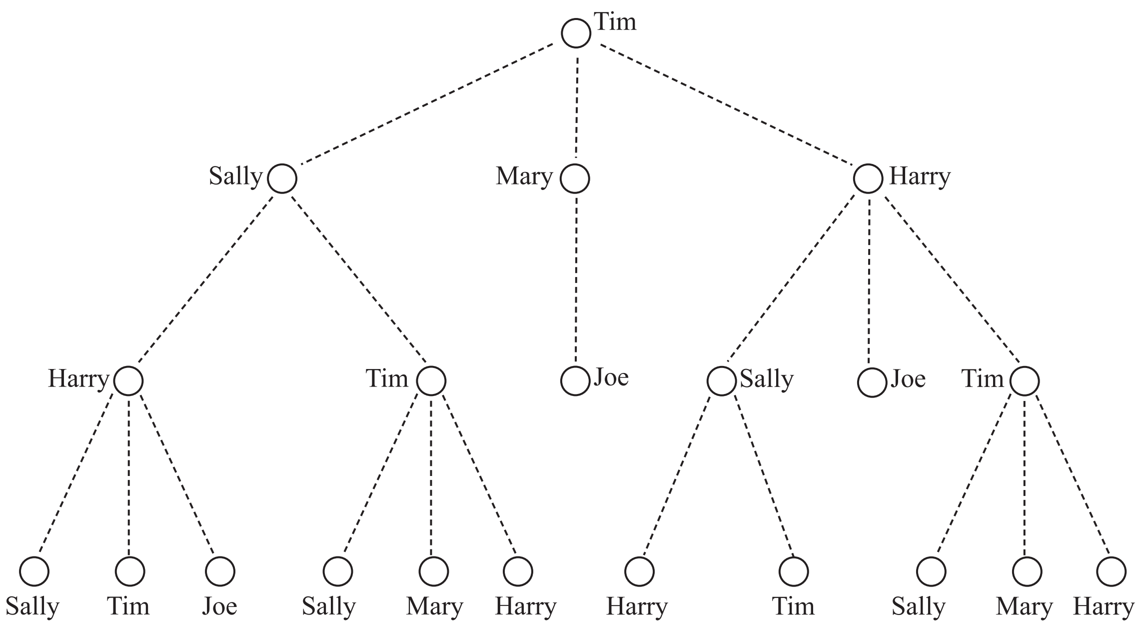

FIGURE 3.7 Walks of length 3 from Tim

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1 connects the two actors (Tim → Sally). Distance may be a useful measure if one is concerned with the quickest route for diffusion, as when, for example, determining the worst-case scenario for an infectious disease to spread in a network.

Walk – any sequence of nodes and edges (can revisit the same node more than once) – for example, Tim, Harry, Tim, Mary
Path – a sequence of nodes and edges starting with one node and ending with another (one can never go back or revisit the same node twice) – for example, Sally, Tim, Mary, Joe
Cycle – A path that starts and ends on the the same node (Harry, Tim, Sally, Harry).
(Geodesic) Distance – number of edges on the *shortest* path between points (Sally to Joe = 2)
Density – number of ties/total possible = $9/(5\times(5-1)) = .45$. Total possible ties is $\frac{1}{2}\ ((n \times n)-n)$ when ties are undirected.

3.5 FROM PICTURES TO NUMBERS

Our first impression of a network is often visual. Network diagrams offer appealing, intuitive representations of the underlying network data. Diagrams make it easy to get a sense of the groups, the regions of cohesion, and the overall structure of a network. A network picture can thus go a long way in telling the story of a social system. And yet, visualizations can be intentionally or unintentionally misleading. Visualizations are also not amenable to numerical manipulation. One cannot add, subtract, multiply, or directly summarize the picture. It is difficult to compare two networks just by looking at pictures. Thus, one needs an analytically tractable mathematical representation of the network data, allowing for numerical summaries of the properties of our network. Note, however, that the visual and mathematical formalizations of network data are tightly connected.

Network data can be formally represented in a number of ways. We begin by describing the *adjacency matrix*, given that it offers the most mathematically direct connection to visual representations. An adjacency matrix represents objects in rows and columns, and the cells are their relations. In the case of one-mode networks, the rows and the columns represent the same set of nodes (say, students) and are arranged in the same order. Formally, one defines a matrix $\mathbf{X}_{ij}$ as an $n \times n$ matrix, such that the elements of the matrix, ij, represent whether a tie exists between i and j. Thus, $\mathbf{X}_{ij}$ represents the value of the tie from i to j. Information contained in the matrix represents the adjacent pairs of all nodes. Row i of the matrix represents the ties sent by actor i, and column j represents the ties received by actor j. The number of actors in the network are represented by n, and thus the number of rows and columns will be equal to n. Note that the matrix will be square, and the order of the rows is the same as the order of the columns.

Table 3.1 presents the adjacency matrix for the Florentine family network plotted in Figure 3.4. The cell entries off the diagonal indicate whether a tie exists between family i and family j: 1 represents the presence of the tie (in this case, a marriage tie), and 0 represents its absence. For example, $X_{9,14} = 1$, indicating that the Medici family and the Salviati family have marriage relations; and $X_{9,6} = 0$, indicating that there is no relation between the Medici family and the Ginori family. Note that the matrix is binary and does not reflect any indication of tie strength. The matrix is also symmetric: the upper triangle is a mirror reflection of the lower triangle. Formally, all $X_{ij} = X_{ji}$ because the network is undirected. When ties are directed and asymmetric, the upper and lower triangles of the matrix will differ unless there is perfect *reciprocity* in the ties. In some cases, the diagonal is meaningful, such as in the case of reported self-liking. In the present case, however, the diagonal is meaningless because it represents self-looping ties of marriages within families. In this period, such ties were a social taboo and did not occur, but the diagonal could potentially have salience for a different era of elites, like Roman Caesars.

This matrix is based on a binary network, but it is possible to capture valued relations within a matrix. Thus, rather than a matrix of 0s and 1s, the matrix would be filled with values describing the relationship between person i and person j. This could be a count (e.g., representing how many times a person sought advice) or a numerical value (e.g., representing how much a person relies on offered advice), but the key is that the data are no longer restricted to 0s and 1s, and this is reflected in the values of the matrix.

We now turn to the matrix representation of the bipartite network introduced earlier (see the matrix in Table 3.2). The data there concern southern women and the social events they attended (see the network in Figure 3.5). The network therefore consists of two types of nodes. When the network is expressed in matrix form, one mode is arranged as rows, and the other mode is arranged as columns. Although the choice is largely arbitrary, it is often easier to select the smaller set of nodes as the columns and the larger set as the rows. In our case, this choice produces a matrix with eighteen rows (women) and fourteen columns (events). Again, we are illustrating a case in which ties are binary. Because the network's nodes are not directly tied with one another but instead are associated or affiliated with one another, the matrix is called an *affiliation matrix*. More formally, the data are represented as an $n \times m$ matrix, where n lists the different actors (i.e., southern women), and m lists the different affiliations (i.e., social events). Element X_{ij} in the matrix is equal to 1 if person i is a member of affiliation j, and 0 otherwise. The affiliation matrix is presented in Table 3.2. Women are shown as the rows; social events are shown as the columns; and a cell entry of 1 indicates that a woman attended a specific event, and 0 indicates that she did not.

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TABLE 3.1 *Adjacency matrix of Florentine family relations*

	Acci	Albi	Barb	Bisc	Cast	Gino	Guad	Lamb	Medi	Pazzi	Peru	Pucci	Rido	Salv	Stro	Torn
Acci	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
Albi	0	0	0	0	0	1	1	0	1	0	0	0	0	0	0	0
Barb	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0
Bisc	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0
Cast	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	0
Gino	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Guad	0	1	0	1	0	0	0	1	0	0	0	0	0	0	0	1
Lamb	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
Medi	1	1	1	0	0	0	0	0	0	0	0	0	1	1	0	1
Pazzi	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
Peru	0	0	0	1	1	0	0	0	0	0	0	0	0	0	1	0
Pucci	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Rido	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1
Salv	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0
Stro	0	0	0	1	1	0	0	0	0	0	1	0	1	0	0	0
Torn	0	0	0	0	0	0	1	0	1	0	0	0	1	0	0	0

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TABLE 3.2 *Affiliation matrix of southern women data*

	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	E11	E12	E13	E14
EVELYN	1	1	1	1	1	1	0	1	1	0	0	0	0	0
LAURA	1	1	1	0	1	1	1	1	0	0	0	0	0	0
THERESA	0	1	1	1	1	1	1	1	1	0	0	0	0	0
BRENDA	1	0	1	1	1	1	1	1	0	0	0	0	0	0
CHARLOTTE	0	0	1	1	1	0	1	0	0	0	0	0	0	0
FRANCES	0	0	1	0	1	1	0	1	0	0	0	0	0	0
ELEANOR	0	0	0	0	1	1	1	1	0	0	0	0	0	0
PEARL	0	0	0	0	0	1	0	1	1	0	0	0	0	0
RUTH	0	0	0	0	1	0	1	1	1	0	0	0	0	0
VERNE	0	0	0	0	0	0	1	1	1	0	0	1	0	0
MYRNA	0	0	0	0	0	0	0	1	1	1	0	1	0	0
KATHERINE	0	0	0	0	0	0	0	1	1	1	0	1	1	1
SYLVIA	0	0	0	0	0	0	1	1	1	1	0	1	1	1
MORA	0	0	0	0	0	1	1	0	1	1	1	1	1	1
HELEN	0	0	0	0	0	0	1	1	0	1	1	1	0	0
DOROTHY	0	0	0	0	0	0	0	1	1	0	0	0	0	0
OLIVIA	0	0	0	0	0	0	0	0	1	0	1	0	0	0
FLORA	0	0	0	0	0	0	0	0	1	0	1	0	0	0

The drawback of a matrix approach is that the matrix is less intuitive than a picture. It is difficult to see emergent social structure, the divisions in the population, or the overall topology with a matrix, even though it can be used to measure these things. Another practical disadvantage of a matrix is that it is a rather expensive way to hold data. Matrices contain n^2 pieces of information (although the diagonal is generally not meaningful because a self-tie is not possible in most cases), representing whether a tie exists between every pair of actors in the network. Because a matrix holds information about non-ties (or the 0s), the amount of information that needs to be held increases with the square of the number of nodes: a network of 10 nodes implies a matrix of 100 elements, but a network of 1,000 nodes implies a matrix of 1 million elements. This feature makes a matrix a poor way to analyze and hold data for very large networks, which is especially noteworthy given the increasing availability of large networks collected from the Internet and digital records.

Alternative means of representing and storing network data are useful. An *edgelist*, for example, is more compact than a matrix but still maintains the same basic information. Each row in an edgelist represents a tie from node i to node j. If the graph is directed, one column (generally the first) captures the *sender* of that tie, and the other column captures the *receiver* of that tie. If the graph is undirected, there is no distinction between sender and receiver. The nodes are identified by an ID or observation number. The edgelist contains all the connectivity information of the figure or adjacency matrix, as every tie in the figure has a corresponding entry in the adjacency matrix and row in the edgelist. The key difference between the edgelist and the adjacency matrix is that the matrix holds a spot for every possible tie in the network, whether it exists or not, whereas an edgelist represents only the ties that are present. The pairs of people not present in the edgelist do not have a tie. The total number of rows in the edgelist is therefore equal to the number of ties in the network.

The edgelist for the Florentine families is shown in Table 3.3. Note that in this formulation, the Pucci family is no longer present because they are an *isolate*, or a node with no connections to other nodes. When using edgelists, one needs to ensure that such isolates are not dropped from the analysis, usually by linking explicitly to a data set that includes all of the nodes. One can also include self-loops to retain cases (i.e., add a looping tie from node i to itself) in the edgelist.

Different network types can be represented using an edgelist. Extending an edgelist to capture valued relationships is simple, requiring only an additional column that contains the value or weights corresponding to that edge. Similarly, we can extend to multi-relational networks by simply including a flag for each relation ("business" or "marriage" in the Florentine example). Affiliation networks, like that shown in Figure 3.5, can also be represented as an edgelist. The first column displays the actors in one mode (in our case, women), and the second column displays the other mode (i.e., events). Each row corresponds to a distinct actor–group tie (e.g., Ruth–Event 6; Ruth–Event 8, and so on).

TABLE 3.3 *Edgelist of Florentine family relations*

NODE1	NODE2
ACCIAIUOLI	MEDICI
ALBIZZI	GINORI
ALBIZZI	GUADAGNI
ALBIZZI	MEDICI
BARBADORI	CASTELLANI
BARBADORI	MEDICI
BISCHERI	GUADAGNI
BISCHERI	PERUZZI
BISCHERI	STROZZI
CASTELLANI	PERUZZI
CASTELLANI	STROZZI
GUADAGNI	LAMBERTESCHI
GUADAGNI	TORNABUONI
MEDICI	RIDOLFI
MEDICI	SALVIATI
MEDICI	TORNABUONI
PAZZI	SALVIATI
PERUZZI	STROZZI
RIDOLFI	STROZZI
RIDOLFI	TORNABUONI

The second common alternative to adjacency matrices in organizing network data is an adjacency list. In an adjacency list, each node is listed as a row and each of its adjacent partners is listed as a separate column. This is the natural way to organize data from a survey in which respondents are asked to name friends, for example. One would then have a row for each respondent and a column for each friend they named. To continue our Florentine example, Table 3.4 includes the same data in an adjacency list format.

Notice here that "PUCCI" is included again, but with no relation named. For an adjacency list, one needs as many columns as the maximum degree in your network (or a limit on alters that a respondent is allowed to name in a survey). If degree is highly skewed, this can lead to a fairly inefficient data structure, as many people will have a lot of empty cells. This data format is common in graph algorithms, as it greatly facilitates exploring paths, particularly when you replace IDs with row numbers. For example, if you want to get all people within two steps of "STROZZI," one need only go to the rows of "MEDICI" and "PAZZI" and see who those people are (ACCIAIUOLI, ALBIZZI, BARBADORI, RIDOLFI, SALVIATI(2), TORNABUONI). For this reason, one typically keeps both sides of an undirected graph in the adjacency list. To capture weighted relations, one needs to construct a parallel data structure with "Alter1weight," "Alter2weight," and so forth.

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TABLE 3.4 *Adjacency list of Florentine family relations*

Respondent	Alter1	Alter2	Alter3	Alter4	Alter5	Alter6
ACCIAIUOLI	MEDICI					
ALBIZZI	GINORI	GUADAGNI	MEDICI			
BARBADORI	CASTELLANI	MEDICI				
BISCHERI	GUADAGNI	PERUZZI	STROZZI			
CASTELLANI	BARBADORI	PERUZZI	STROZZI			
GINORI	ALBIZZI					
GUADAGNI	ALBIZZI	BISCHERI	LAMBERTESCHI	TORNABUONI		
LAMBERTESCHI	GUADAGNI					
MEDICI	ACCIAIUOLI	ALBIZZI	BARBADORI	RIDOLFI	SALVIATI	TORNABUONI
PAZZI	SALVIATI					
PERUZZI	BISCHERI	CASTELLANI	STROZZI			
PUCCI						
RIDOLFI	MEDICI	STROZZI	TORNABUONI			
SALVIATI	MEDICI	PAZZI				
STROZZI	BISCHERI	CASTELLANI	PERUZZI	RIDOLFI		
TORNABUONI	GUADAGNI	MEDICI	RIDOLFI			

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These three data structures – adjacency matrices, edgelists, and adjacency lists – are the most common ways to organize social network data, but we will note that there is a class of network data formats that you may run into based on different sorts of "markup" strategies, where the data are stored much like an HTML file. These files include tags that tell a program what each element in the file means, and the file is generally nested to describe different levels of data. So, for example, you may have lines in the file that describe the graph as a whole, then lines that describe nodes and all their associated data, and then lines that describe relations and all their associated data. For example, here is a brief snippet of an XML representation of the Florentine data:

```
<graph id="G" edgedefault="undirected" parse.nodes="16"
parse.edges="20">
  <node id="v1"> <data key="a1">ACCIAIUOLI</data>
   <data key="a2">0.1028396</data>
   <data key="a3">0.6620292</data>
   <data key="a4">0.5000000</data>
  </node>
  <node id="v2"> <data key="a1">ALBIZZI</data>
   <data key="a2">0.5797447</data>
   <data key="a3">0.5207278</data>
   <data key="a4">0.5000000</data>
  </node>
...lines cut...
  <edge source="v1" target="v9"/>
  <edge source="v2" target="v6"/>
  <edge source="v2" target="v7"/>
```

Note that the version above used 153 lines of code to describe 20 marriages among 16 people, so it is not a particularly efficient format! However, there are many variants of this format that you will run into because it is an easy way for machines to read and write such files while still being somewhat readable by humans and avoiding many of the problems of proprietary binary formats (of which there are also many) (see Tsvetovat, Reminga, & Carley 2004 for discussion of attempts to unify such data representations). These are often the default data formats for data produced by application programming interfaces (APIs) for Twitter or GitHub network data.

Thankfully, most users rarely have reason to interact with most of these data structures directly. The tools used in R – mainly, the *statnet* and *igraph* packages – each have their own network data formats (different types of network "objects"), each of which hold data just a little differently. Once one has moved data out of a source file (Qualtrics survey result or Twitter API feed) and pulled them into R, these issues of data structure become largely irrelevant. What is important, however, is for users to know enough about the data

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structure to avoid making silly data management mistakes – by mismerging data, for example. As such, we believe it is important to have a basic understanding of how data are held, organized, and managed.

3.6 SUMMARY

We have progressed from a general informed understanding of social structure (Chapter 2) toward the empirical science that is social network analysis. Networks are abstractions built around visual and mathematical representations. In this chapter, we have introduced the reader to the key building blocks of network analysis: nodes, ties, and matrices. We are well on our way from moving from the reality of social structure to knowledge (useful information). And yet, to move forward, we must first have the sort of data that can be represented through the kinds of abstractions presented in this chapter. And if those data have not already been collected, we have to go out and get them.

SUGGESTED FURTHER READING

In addition to the general methods texts listed for Chapter 1, useful deep dives into methods discussed here include:

Gibbons, Alan. 1985. *Algorithmic Graph Theory*. New York: Cambridge University Press. (A classic exposition of primary algorithmic approaches to manipulating, searching, and performing search operations on network objects. While some of the routines have been superseded by newer algorithms, this publication provides basics to get anyone started in this line of work.)

Harary, Frank. 1969. *Graph Theory*. Reading, MA: Addison Wesley. (The classic work in the basics of graph theory.)

Kivelä, Mikko, Alex Arenas, Marc Barthelemy et al. 2014. "Multilayer Networks." *Journal of Complex Networks* 2: 203–71. (Multilayer networks are one of the most general ways to bind multiple types of nodes and relations in a single analysis object; the approach has proven useful for both cross-context studies and unique approaches to dynamic modeling of networks.)

Richards, William, and Andrew Seary. 2000. "Eigen Analysis of Networks." *Journal of Social Structure* 1(2): 1–17. (Eigen structures are key to many network metrics, although their use and mathematical foundations are often understudied in social sciences. This work provides a gentle introduction to what they are and why they matter.)

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4

How Are Social Network Data Collected?

We outline key conceptual issues and strategies in social network data collection, focusing on the differences between realist and nominalist approaches. Given that most networks are incomplete in some way, we discuss ways to anticipate and assess problems with missing data.

4.1 SOURCES OF NETWORK DATA

Now that you have a sense of what networks are and why to study them, the next step is to consider how to collect and analyze network data. Network data collection strategies take many forms, including small group questionnaires, large-scale surveys, face-to-face observations, "scraping" many thousands of websites, APIs, or digital archives. The choice of strategy is informed by the research focus and the available data sources. Many studies use previously collected network data, and the same caveat holds: it is crucial that the problem at hand is suited to data. As we have already discussed, conceptualization and theorization enter even as one begins to collect data.

The most empirically valid studies consider node sets that are clearly bounded by space and time. One of the earliest network studies, of the Hawthorne Plant's telephone bank wiring room, examined the multiple relations among workers and supervisors in a single room (Roethlisberger, Dickson, & Wright 1947 [1939]). Clearly, a single room in a plant is not a complete network: these individuals likely had many friendships outside that room, even at the same plant. However, because the outcome of interest for the research team concerned work productivity, the flows of interpersonal influences that were most likely to bear on this outcome were those in the immediate work environment. Many network studies will face a more difficult task in defining the boundary of the network (i.e., defining the set of nodes that are part of the study). For example, researchers interested in how intellectual

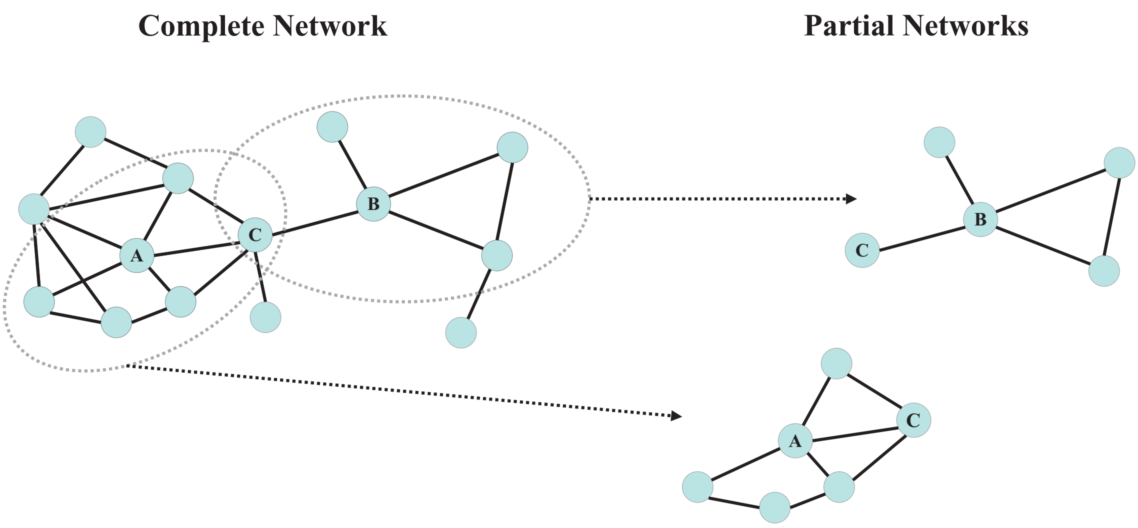

FIGURE 4.1 The difference between complete and partial network data

influence flows through academic research communities must decide where to draw reasonable boundaries among thousands of individuals. As always, this process points to the necessity of a theoretical and substantive understanding of the field. It is really only on these grounds that the researcher can justify the choice of a node set.

While most networks necessarily are partial views of social structures, it is useful to distinguish between local and global data collection strategies, as illustrated in Figure 4.1. Local networks are also called *egocentric* or *partial* networks because they capture a local structure of ties centered on a particular node, an *ego*. Essentially, individuals create and are at the center of their own local network. As we cover in Chapter 6, such networks yield potentially insightful information about individuals and their social ties, but these are not complete networks in the sense of being a globally interconnected system among all the nodes in a study. In contrast, global networks are not centered on a single individual but instead on a larger group. Global networks are therefore referred to as *sociocentric* or *complete* networks because they capture a full census of the nodes and relationships within a particular domain. Figure 4.1 shows how the partial network egos A and B do not correctly indicate the boundary-spanning position of individual C, which would be found in a complete network. Although global networks provide a more complete structural picture, one should not exaggerate the differences between local networks and complete networks: both are subject to practical data collection problems. Regardless of the type of data used, the researcher must be aware of issues of missing data and potential biases, as well as the difficult task of defining the boundary of the network.

Gathering local and global network data involves different sampling strategies, and these strategies present both benefits and drawbacks. Indeed, local network data have several strengths. Local network structures are generally

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based on random probability samples; thus, ego network data are easy to collect and offer a (potentially) wide coverage of the population. One could, for example, collect egocentric network data on a random sample of the entire United States, as the General Social Survey (GSS) has done (Burt 1984; Marsden, Smith, & Hout 2020). Similarly, egocentric network data make it easy to capture a wide range of network *alters*, defined as those people that an ego names as a social tie. The named alters need not be limited to certain geographic regions or to certain types of people (e.g., only those in a given organization), reducing the problem of setting boundaries in the study. Egocentric network data are sufficient to capture many personally relevant network properties, such as access to social support and homogeneity of personal networks (e.g., whether an ego generally knows people with similar levels of education). Ego-level network data are also useful in measuring certain global network properties. When the outcomes of interest concern population-level indicators of, say, racial segregation in friendship ties, then local network data are sufficient to draw unbiased inferences to the global structure.

Ego network data are, however, ultimately local in nature – that is, egocentric data are embedded in sociocentric data, but the reverse does not hold. This property makes it difficult to measure global network properties, such as walks and paths, which are dependent on the pattern of interaction between all actors. Additionally, ego network data often capture subjective understandings of relationships and miss the relational perceptions of alters. As such, they are perceived ties rather than externally confirmed ones, and they could reflect wishful thinking in some cases. Because actors are almost always embedded in a wider array of ties, it is important to move beyond the individual pieces afforded by ego network data to say something about the features of the whole network. Recent work has made great strides in solving this problem, using statistical models to construct complete networks based on the sampled data (e.g., Krivitsky & Morris 2017; Smith 2012; Smith & Burow 2020). Thus, a researcher could use independently sampled network data and still infer the features of the global network, combining the convenience of ego network data with the holistic feel of sociocentric data. Network inference remains a difficult problem, however, and is an open area of research.

Global network data that are not inferred through a large number of local networks are gathered by painstakingly asking all the nodes within a specified boundary to report their ties to all others within that boundary. Consequently, global network data are costlier to collect. Researchers trying to decide on a local versus global network study design should weigh the costs and benefits of each. For example, the ease/difficulty of data collection (local network data are easier) should be weighed against ease/difficulty of measuring network properties (global network data are easier). That said, when collecting complete network data is possible, it is often the preferred route.

Whether local or global network data are collected, research design involves making decisions concerning two definitional issues: (1) defining the boundary of the group from which the node set is drawn and (2) defining what constitutes a tie among group members. Researchers tend to resolve these issues using either a more *realist* set of approaches in which participants are allowed to more inductively generate the members of the group and ties among them, or more *nominalist* approaches in which the researcher more deductively determines the members of the group and the existence of ties (Laumann, Marsden, & Prensky 1983).

4.2 REALIST AND NOMINALIST APPROACHES TO BOUNDARY DEFINITION

Realist approaches are sometimes called *emic*, because they rely on internal definitions and perceptions. Thus, a network realist lets subjects define boundaries. Using a name generator within a survey is one realist approach to collecting local network data. A name generator approach allows the respondents to produce a set of names that are included in their egocentric network, as well as their perceptions of ties among these individuals. The respondents thus define the boundary (or the inclusion of nodes) in their ego network. Frequently, the researcher asks respondents to list all the individuals with whom they have had a certain relation in the recent past. A common format focuses on contacts with whom an ego has discussed important matters in the last six months. Alternatively, a researcher may ask about friendship, social support, or even about specific exchanges (e.g., identifying who the respondent could ask for a small loan). A realist approach may also extend beyond defining the boundary of network participants to include what types of ties are relevant within that context. Do people regularly seek advice for getting work done within a given firm? How important are formal collaborations in establishing one's status in this workplace? Knowledge of what ties matter in a given context are key to a realist's approach to validly establishing where a network begins and ends.

In general, it is often beneficial to ask about multiple relationships within the survey because the respondent's friends may or may not be the same people they seek advice from (using different "name generators"). Afterward, the respondent is asked to work through that list of close contacts to supply information – such as gender, education, and occupation – about each one ("name interpreter" questions). Finally, the respondent may be asked to provide further network information for each pair of listed contacts, such as whether person *j* is close to person *k*. Obviously, this can be a time-consuming procedure, and respondents may fatigue and provide progressively less reliable data as they complete the task. To reduce this possibility, researchers will frequently limit the number of names generated. For example, a respondent may be asked to list up to five individuals. Of course, this comes with its own trade-off of underreporting the true number of close contacts for an unknown number of

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respondents. On the other hand, placing no limits on the number of contacts risks that some respondents will regard everyone as a close contact – an impossibility given that actual close ties have costs and require time and effort (Jackson 2008). The goal is to find an adequate range that reasonably applies to most people and captures variation. If most respondents fill in five ties, the number is probably not set high enough. If the number is set to ten, and 10 percent report that many, then the number is more reasonable. For other types of ties, different issues will need to be considered, but the point remains: be reflective and critical of your data collection strategy.

Realist approaches that seek more globally complete networks tend to use one of several techniques to derive snowball samples of nodes. Here, the researcher starts with a name generator, as with more local network approaches, but then gathers contact information for the people named by the first respondents, who are in turn contacted and asked to provide further contact information. This continues until some cutoff criterion is met – for example, when there is some indication that the global network is saturated, and the listed actors and ties become redundant. It can be difficult to determine when to stop a snowball sample, especially when the sample is dependent on the starting seeds or the actual structure is quite segmented (Mouw & Verdery 2012). In practice, studies are often shaped by practical concerns and stop after a certain number of respondents have been interviewed. Thus, although the goal is to gather a network that is as complete as possible, networks gathered based on snowball samples (and similar techniques) are often incomplete.

In contrast to realist approaches, researchers adopting nominalist techniques define the group boundary before collecting data, based on shared social positions or shared events. Because nominalists rely on external sources and judgments, the approach is sometimes called *etic*. Based on shared position, the researcher might define the network boundary as all teachers at a particular school or all employees at a midsize company. Based on shared events, a network could include the regulars at a country club or the active members of a church.

For complete network data, researchers often have to adopt a nominalist boundary on practical grounds. This is especially the case for digital and company records, where information on external relationships is hard to acquire. In those studies, the focus is on relationships internal to a website or a firm, and the assumption is that external influences and unrecorded instances of association are significantly less important for predicting firm and participant outcomes than those persisting within the nominal boundary as defined by the researcher, website, or firm. However, these two boundaries – realist and nominalist – are often more aligned than orthogonal, especially under certain conditions. For example, in demanding institutions, such as American high schools, youth direct their attention more inward toward their peers than outward, say, to persons in their workplaces – that is, they favor selecting in-school peers as their friends over out-of-school peers and adults (Coleman

1961). In this instance, a nominalist school boundary is often one that is also a realist boundary as defined by the students themselves. However, this convergence breaks down when students are asked whether they befriend adults at school or, in some schools (e.g., commuter schools), whether they actually form primary relations outside of school.

4.3 REALIST AND NOMINALIST APPROACHES TO TIE DEFINITION

Researchers can also use realist or nominalist approaches in establishing the existence of a relation. The aforementioned name generator is a good example of a realist approach that defines both the boundary and the ties using a respondent's subjective experience. A researcher may also take a more nominalist approach in defining the boundary of the network but then allow respondents to determine the ties that exist, thus engaging in a constrained realist approach to determining relations. For example, the individuals deemed part of the group of interest (e.g., school or neighborhood) would be contacted and given a questionnaire with a roster of all other individuals in that setting. The respondents would then indicate which people on the roster they have relations with (advice, friendship, mentorship, and so on). Of course, the researcher may not have such a complete roster and may rely on respondents recalling their relations (with sufficient identifiers to map out the network). This situation involves a trade-off: a roster may elicit more false positives (e.g., wishful thinking), but the recall method is likely to garner more false negatives (e.g., forgetful thinking).

Network data used to be gathered with written questionnaires, such as the one in Table 4.1. Today, the same basic information is typically gathered with online or tablet-based survey platforms that ease integration of data collection and analysis (Birkett et al. 2021). Nonetheless, one will generally use a roster of some kind or match responses *post hoc* to generate the network from such data.[1]

A special case of the realist approach is collecting *cognitive network* data in which the researcher asks respondents to report on the relations they believe each individual has to others in the group. Respondents report not only on their own perceived ties but also on the ties they believe to exist between other actors in the network. For example, each member of a high school classroom might be asked to draw a network of all the friendship ties among their classmates, providing a comprehensive view of the social structure from each participant's point of view. Such cognitive network information can be used for a variety of

[1] This choice often represents a trade-off in researcher versus respondent effort. For example, it is fast to ask students in a classroom to write down the names of their friends, which minimizes the in-class time students have to spend on an assessment. However, this sort of open-ended naming results in a significant name-matching effort on the part of the researcher needing to disambiguate named peers (see Moody et al. 2010 for discussion).

TABLE 4.1 *Sample network questionnaire*

Friends Nomination Form – **Who are your close friends that you usually hang around with? Please list only as many people as you usually hang out with.**

1.	**2.**	**3.**	**4.**	**5. In what settings do you usually see this friend?** *For each friend check as many as apply*							**6. When do you see this friend?** *Check as many as apply*			**7.**	**8.**
What are your friends' full names? *Please print their first and last names*	About how old is this friend?	How long have you been friends?	Is this friend male or female? *Check Male or Female*	In my school classes	In a school activity (like a team or extra-curricular)	In a non-school club or activity (like a youth group, or church)	At work	In my neighborhood	In my family	Other	Less than Once a week	Weekdays	Weekends	Do you know this friend's parents? *Check Yes or No*	Is this friend a best friend? *Check Yes or No*
Example: Jane Doe	16 yr.	6 mos.	___ Male X Female	X	X	X						X	X	X Yes ___ No	___ Yes X No
(a)			___ Male ___ Female											___ Yes ___ No	___ Yes ___ No
(b)			___ Male ___ Female											___ Yes ___ No	___ Yes ___ No
(d)			___ Male ___ Female											___ Yes ___ No	___ Yes ___ No
(e)			___ Male ___ Female											___ Yes ___ No	___ Yes ___ No
(f)			___ Male ___ Female											___ Yes ___ No	___ Yes ___ No

(*continued*)

TABLE 4.1 (*continued*)

Friends Nomination Form – Who are your close friends that you usually hang around with? Please list only as many people as you usually hang out with.							
1.	2.	3.	4.	5. In what settings do you usually see this friend? *For each friend check as many as apply*	6. When do you see this friend? *Check as many as apply*	7.	8.
(g)			_____ Male			_____ Yes	_____ Yes
			_____ Female			_____ No	_____ No
(h)			_____ Male			_____ Yes	_____ Yes
			_____ Female			_____ No	_____ No
(i)			_____ Male			_____ Yes	_____ Yes
			_____ Female			_____ No	_____ No
(j)			_____ Male			_____ Yes	_____ Yes
			_____ Female			_____ No	_____ No

purposes, such as ascertaining which relations are most recognized by all participants in the class. The basic argument is that the most recognized ties are also the ones with the highest level of obligation (see Cairns & Cairns 1991). Another approach to cognitive networks is more laborious and asks respondents to report the relationships they think each different member of the network believes they hold (Krackhardt 1987). For a network of ten people, each person would provide ten sociometric surveys reflecting their sense of how each actor would report their ties! This sort of cognitive network information can be combined to offer a more detailed depiction of a cognitive social structure. From here, one can ascertain who has the most accurate sense of the cognitive social structure, or whose reports on alters' ties best reflect ego-reported connections. When a respondent correctly surmises the ties that others' report, they are said to have an accurate perception of the social structure. These individuals are often considered potential leaders of firms and tend to be central to the network. Conversely, persons who are hard for group members to place in the network are often regarded as enigmas and may not benefit from their own perceived ties.

In contrast, nominalist approaches determine the presence or absence of ties by drawing on interaction patterns observed either in person or through communications technologies; archival records demonstrating proximity or contact; and recorded traces of exchanges of various kinds. In comparison to the more subjective judgments of respondents, nominalist strategies are often able to garner more objectively defensible networks from existing data sources. Nominalist approaches are also likely to scale better than subjective approaches (which require interviews of all actors), making it possible to collect complete network data on large populations. A researcher may have the phone records of an entire country or the email exchanges of a large university. One possible drawback of nominalist approaches is that the data are often limited in terms of the kinds of relationships that are collected. A researcher with email data may not have any other information on the relations between actors, making it difficult to discuss things like friendship and support (or bullying and dislike). Similarly, because the data are automatically collected (or are archival data), information on the nodes in the network may be limited; for example, demographic information may be limited or nonexistent. With such limitations in mind, recent work has begun pairing surveys with nominalist network data, offering a rich data set but also limiting the size of the system that can be studied.

In addition to simply determining the existence of ties, collecting network data often involves indicating the *strength of ties*: the value attributed to the relation in terms of frequency, duration, emotional intensity, and so on (Granovetter 1973; see also Marsden 1990). A realist approach might ask respondents about how frequently they seek professional advice from each of their relations at work or about the extent to which they would characterize a relation as "loving." In an ego network survey, respondents may provide information about the duration, strength, or type of relationship for each alter named.

Tie strength often has an inherently subjective quality because it frequently involves individuals' feelings. Therefore, nominalist approaches to ties may have validity issues when attempting to discern the strength of a tie. Although there are many valued objective indicators of ties – such as the number of emails i sent to j, or the number of articles researchers i and j have coauthored – the concept of tie strength often entails individuals' feelings and perceptions. For example, if the main mechanism thought to transmit a flow of influence is awareness of another node, the researcher using more nominalist indicators of tie strength may mistake *propinquity* (physical proximity) for subjective importance. Having frequent contact is certainly one way of thinking about tie strength, but it falls short of the richness inherent to the concept.

In short, both nominalist and realist approaches are susceptible to measurement error when recording ties, but having both objective and subjective measures may be useful in exploring the substantive question of interest. Given this situation, what can be done? As with all data collection, the researcher must carefully consider issues of data quality and potential sources of bias.

4.4 DATA QUALITY

Bias and error are present in both attributed nominalist and reported realist ties. Both approaches are prone to omitting ties and erroneously reporting them. For nominalist cases, the records used may be loosely entered or partial. The researcher therefore needs to show some reflexivity and openly report the steps taken to address every kind of substantive error. Although one may never achieve perfection, one may find that one gets closer to it over time.

4.4.1 Data Issues in Nominalist Approaches

Take, as a nominalist example, the effort to study researcher collaborations in journals by using Web of Science or Scopus – common data sets from which to derive networks. First, each corpus fails to capture an unbiased sample of scholars. They are biased toward journals written in English and therefore toward English-speaking academics. They also fail to include both "slow" science (e.g., books) and "fast" science (e.g., conference proceedings). Researchers taking a random sample of CVs would find that these corpora are systematically biased against certain kinds of scholars and scholarship. Second, the notion of ties via coauthoring may be erroneous in some cases. Not every collaboration is the same: some collaborations among physicists entail 500+ scholars (e.g., CERN particle accelerator research), and others, like ours, entail four. It would be unrealistic to argue that all 500 physicists share ties like the four of us coauthoring this text do. Supplemental analysis can be used to see, for example, which collaborations are repeated or whether the persons involved adopt one another's citations, thereby discerning which team sizes actually are the most prone to these substantive forms and outcomes of association (Rawlings et al. 2015). From there, one can argue that collaborations of, say, eight or fewer scholars exemplify relatively substantive and persistent bonds. Third, corpora like Scopus cover only one type of tie or activity salient to participants in the academic domain. Scholars also teach, advise students, marshal theses, consult for firms, apply for grants, patent ideas, participate on committees, and work in departments. Presumptions of the independence of one tie from all others introduces concern. Researchers must consider whether they are forgetting a key confounder or other type of association that can influence the understanding of tie effects or social structure formation.

As another example, take Facebook friends. Many students think in terms of such data today, but much can be said about error inherent in those data. First, the representativeness of the sample is questionable. Many Facebook pages are groups or fan pages. Some even are of deceased individuals. One way to build validity into ties gathered online is to focus on a subset of known persons in an organization – say, a single university and its students (see Wimmer & Lewis 2010). Using a roster, one can see how representative the Facebook network is of that university's students. Good coverage would suggest a reasonable sample of those individuals. Without such grounding, it would be impossible to know who these data represent – and 100 years of careful research showing the importance of sampling would be ignored in the process! Second, it is difficult to know whether one person's 1,000 friends on Facebook are the same type as the 100 friends another person lists. Using supplemental information on who posts on whose page, how often they correspond, how often they are tagged together, and so on, one may be able to identify which of these listed friends are friends of different types (Garton, Haythornthwaite, and Wellman 1997; Rainie & Wellman 2012). Then our comparisons may be more aligned. Next, recorded activity on Facebook over a particular period could vary widely across users: one individual may use Facebook 100 times more than another individual, resulting in 100 more observations of their ties than for the other individual. To offset such activity bias, the researcher might cluster users by how recently and frequently they access the platform in a given period and then compare users within those sets, or researchers can use those activity rates as controls to offset sampling bias. In the case of the millions of observations found on social media, these clustering approaches become more and more feasible and recommended.

This sounds like a mountain to overcome, and it is. But the careful researcher will think about it, chip away, and report openly and honestly. Scientific progress depends on building on others, so contending with these issues should be viewed as a matter of good practice to master rather than an outcome (publication) to achieve.

4.4.2 Data Issues in Realist Approaches

Realist approaches to ties have similar concerns to those faced by nominalists. When participants report ties, they may be delusional or report wished-for

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relations. People routinely forget their interactions and relations. They may even report what they think the researcher wants to hear (i.e., social desirability bias). Furthermore, they can be primed to remember certain things or to frame responses in certain ways. In fact, even the definition of a tie may be highly contingent on the cultural background of the respondent: what you call a friend may not be the same as what we call a friend! All this may create an unnatural bias or fail to acquire information acted on in an emic fashion by participants "in the wild."

Various strategies offset bias in tie-reporting. First, a balance of open and guided reporting helps. The goal is to avoid too few reports (false negatives) as well as fatigue and inaccuracy from too many reports (false positives). Aids such as more entries, rosters, and name autocompletion can help with the former issue; limits on entries and options for "other" name versions (nicknames) may help with the latter. Second, the framing of questions is essential (see Bradburn, Sudman, & Wansink 2004; Krosnick 1999). When asking persons to report who their friends are, it helps to define "friend" and establish a time frame. For example, a staple question is, "List up to 10 friends you hung around with this semester outside of class." The words "hung around with this semester outside of class" are key, helping to specify what is meant. Third, attempting to assess the quality or strength of tie can also offset error. Surveys can include a barrage of tie quality questions in checkboxes to specify what a friendship entails. Examples include survey items asking which days of the week they see each other, in what contexts they see each other (in class, school, work, neighborhood), how close they are, how much they trust the friend, and how long they have known each other. Notably, research has found that persons are more consistent in reporting behaviors than sentiments. Using these three strategies can help identify notions of friendship that are comparable and, at the very least, identify ties that meet a certain minimum threshold of reliability.

All this may sound glum. Not at all! The point is that the researcher needs to be empirically vigilant. What is needed is not more and faster science, but instead more thoughtful and careful science. Consider what you are studying, and reflect on the sample and its boundary. Mull over the nature of that social structure and the outcome of interest, consider confounding factors and alternatives, and identify and compare similar entities. Consider combining data sources, if possible, to offset the weaknesses of one with the strengths of the others. As a result of these efforts, the associations identified and inferences performed will be more valid and generalizable. And where you see limitations in your research, report them. This kind of reflexive, deliberate research will produce published work with a longer shelf life, and all who follow will be able to build from and improve on it.

Table 4.2 offers a summary of the trade-offs in realist and nominalist approaches. The reader may find this useful as a quick reference guide to consult when developing a network data collection strategy.

TABLE 4.2 *Comparison of realist and nominalist data collection strategies*

	Realist	**Nominalist**
General	Network features based on subjects' internal definitions and perceptions (*emic*).	Network features based on researcher-defined boundaries and definitions (*etic*).
Boundary	Respondent free to define relevant alters based on perception of question (e.g., the GSS "important matters" question).	Researcher defines relevant setting, as question frame (name in-class friends) or records boundary (all company employees).
Ties	Respondent interprets name generator freely (e.g., no specific anchor to interpret "important matters"). Values interpreted by respondent ("close friend" vs. "acquaintance").	Archival, observed, or recorded traces of interactions and exchange (well-anchored survey responses are middle-ground cases between realist and nominalist). Values summed from observations; linking instances to meaning can be problematic.
Special Features	Cognitive social structure data allow each respondent to provide their own perception of relations among all others.	Nominalist approaches make it simpler to include noncognitive actors via multilayer networks (e.g., linking people, animals, and parasites in a disease study).
Data Issues	People may have widely varying understandings of what relations mean. Common relations are internally multiplex (e.g., "friend" can mean admiration, trust, or respect). Cognitive and instrument biases (recency, proximity, social conformity). Respondent fatigue.	Archival data on interaction traces often provide limited node-level information. Archival/web data may have systematic coverage biases (i.e., politically active youth in Twitter, English in Web of Science). Rosters can easily go out of date in highly mobile settings.

4.5 ISSUE OF MISSING DATA

Even when sample bias is low and measurement error is minimized, a researcher is still likely to face one more key threat to validity: missing data. Missing data are a common but often ignored problem in network studies, although a growing literature addresses such issues (e.g., Borgatti, Carley, &

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Krackhardt 2006; Kossinets 2006; Smith & Moody 2013; Wang et al. 2012). Missing data imply that a researcher has incomplete information about the population of interest. With regard to network analysis measures, which by assumption depend on connections among all actors, missing information makes it difficult to map out the full pattern of connections, thereby leading to potential biases in the measures of interest.

Missing data can take a number of forms. For example, a study may have missing nodes: there is no information on a subset of actors in the network, even though all actors in the population have had a chance to participate (making it not a boundary problem per se). Nodes may be missing for a variety of reasons, depending on the nature of the study. For example, in a traditional survey in a school, some students may be absent the day of the survey. Similarly, a study that uses cell phones to track interactions (either via calls or proximity) may have missing nodes for participants who lose their phones or opt out of the study (e.g., by turning off Bluetooth). Analogous examples can be imagined for nearly any population of interest. The consequences of missing nodes are shown in Figure 4.2. Panel (a) plots a simple, complete network with

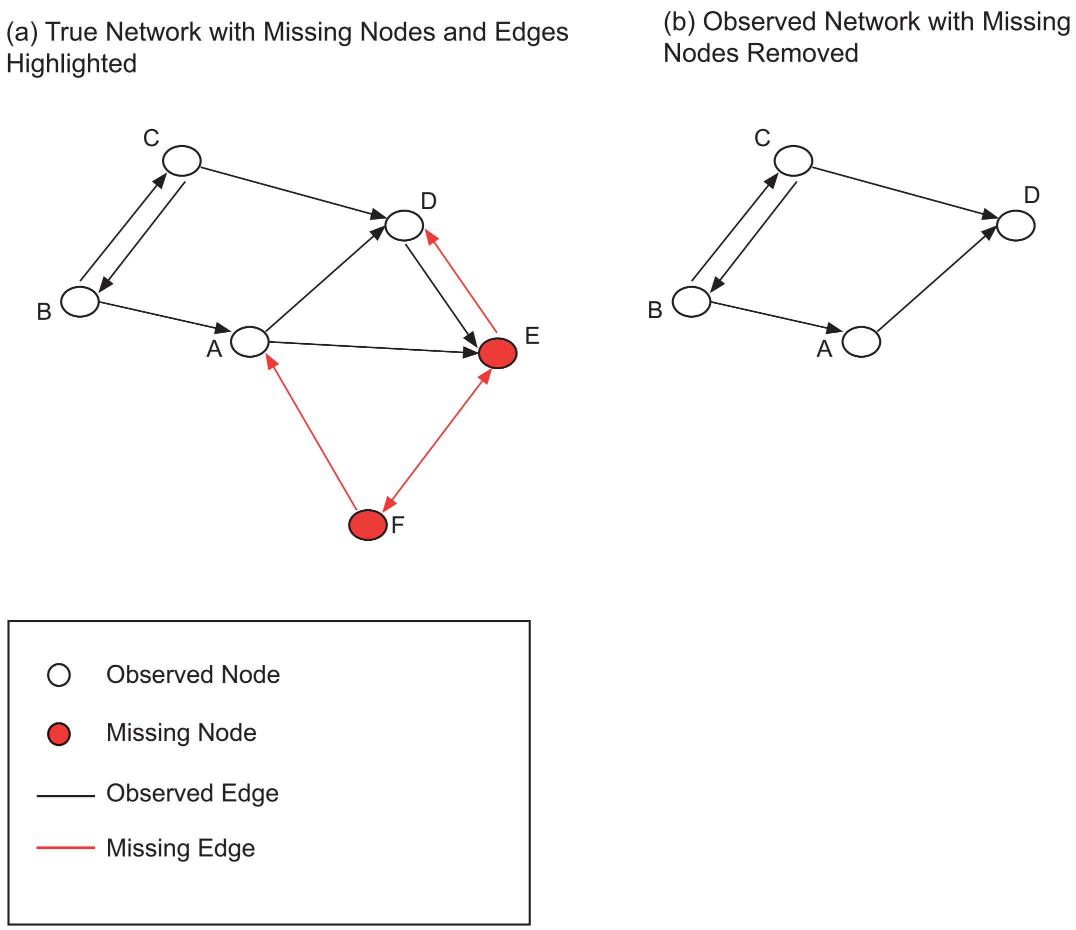

FIGURE 4.2 The consequences of missing network data

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two example missing nodes highlighted in red. Panel (b) shows the same network with the missing nodes removed. Clearly, the network with missing data offers an incomplete picture of the true (unobserved) network in panel (a).

Past work documenting the deleterious effects of missing data on network measurement (e.g., Costenbader & Valente 2003; Smith, Moody, & Morgan 2017; Wang et al. 2012) shows, of course, that the level of bias increases as the level of missing data increases. But more important is the great variation in the effect of missing data on network measures. A study with the same level of missing data may end up with quite high or quite low amounts of bias, depending on the confluence of a number of factors, including the type of missing nodes, the measure of interest, and the features of the network in question. For example, missing more central nodes generally leads to more bias, but some measures and network types are particularly susceptible to missing important nodes (e.g., certain centrality measures in a small network, and almost any measures reliant on path distances). In general, missing data will be particularly important in networks of a small to medium range, where missing particular nodes has a greater chance of affecting the network structure.

4.6 Anticipating and Accounting for Measurement Error and Missing Data

Ultimately, all studies have limitations, and a researcher should take proper care to identify, assess, and minimize any clear sources of bias. Two basic strategies can be used to address threats to validity, and ideally these two would be used together to form a robust strategy. First, a researcher may try to *adjust and/or impute* the raw data in some way to account for biases. Second, a researcher may examine the *sensitivity* of the estimates and substantive conclusions to different validity threats.

4.6.1 Adjusting and Imputing

A researcher faced with measurement error and missing data can adjust the raw data in a number of ways, depending on the problem at hand. For example, sample bias can be corrected for if the actual population of each group is known by other means. Then one can apply sampling weights (the inverse of the likelihood of being sampled) to bring the observed data back in line with the population of interest. This procedure can apply well to groups, but it is less clear how to apply it to relationships.

For missing edges, traditional statistical solutions cannot be easily applied. What should a network researcher do in the face of missing data? Listwise deletion (where all nodes with missing information are removed from the network) is always an option (and a common choice), but this is not

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particularly attractive, given the potential bias introduced in networks with missing data. Alternatively, a researcher could attempt to impute the missing ties. Imputation strategies range from quite simple to complex, and the relative returns (in different research settings) of different imputation methods is still very much an open question.

With "simple imputation," a researcher uses the nomination information from the observed nodes to the missing nodes to reconstruct the network. Such an imputation process is based on the assumption that the researcher has information about the ties going to the missing nodes, or nonrespondents. For example, in an organization, a worker may be out sick on the day of the survey but may still be on the roster such that other members of the organization could nominate them. When respondents nominate someone who does not complete a survey, that information can be used to impute missing replies and tie affirmations. For example, as shown in panel (a) of Figure 4.3, nodes A and

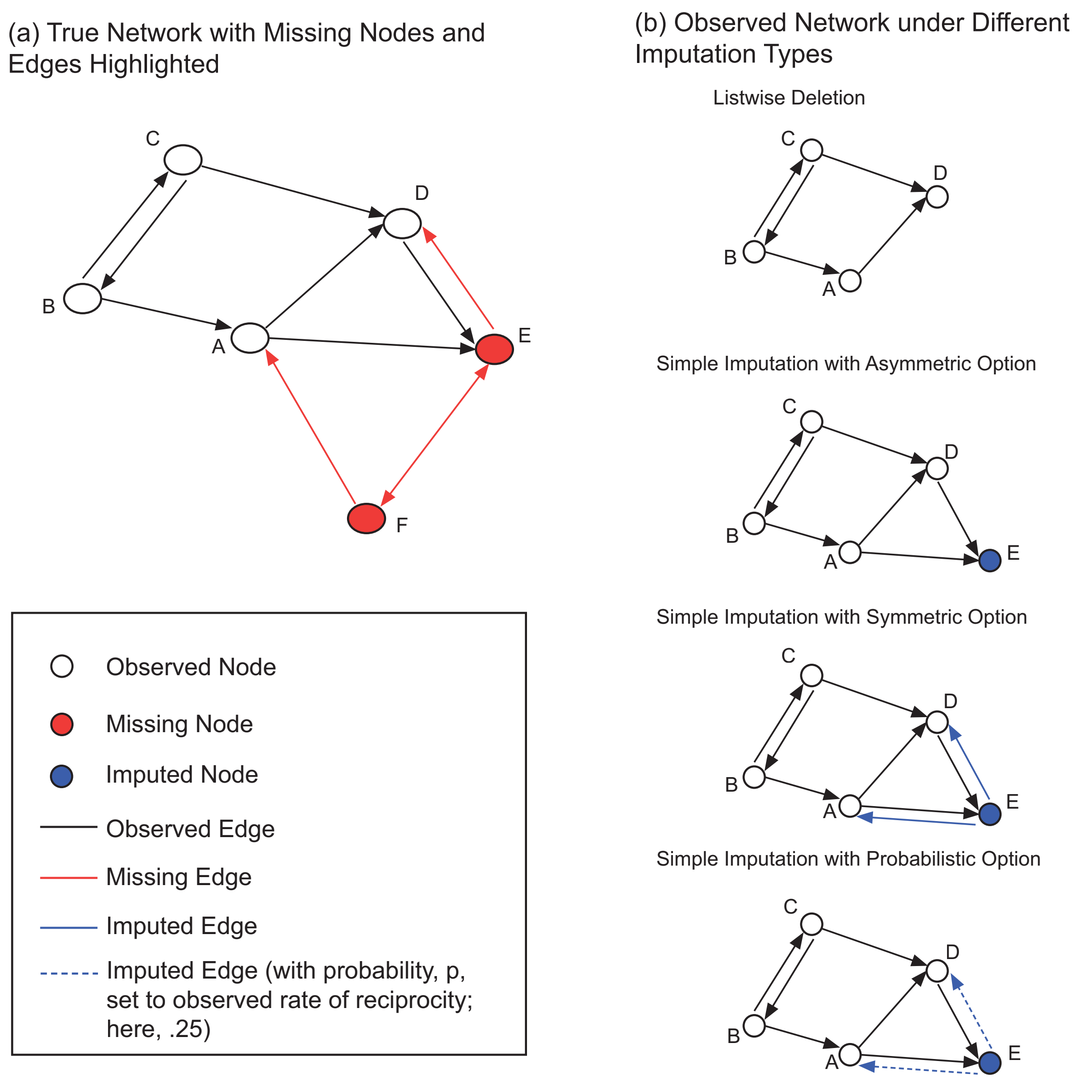

FIGURE 4.3 Simple imputation options for missing network data

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D nominate node E, who is missing from the network. As an imputation strategy, one could put node E back into the network with edges from A to E and D to E. We demonstrate this in the second network depicted in panel (b) of Figure 4.3. Note that node F does not get put back into the network because there are no ties from respondents to F, the missing node.

A researcher must then decide on how to handle ties from nonrespondents to respondents who nominated them. In this case, one needs to impute whether E sends ties back to A and D. One option is to simply assume that all ties from nonrespondents to respondents who nominated them are reciprocated; thus E–D and E–A would be imputed, as depicted in the third network in panel (b) of Figure 4.3. Alternatively, one could impute the ties probabilistically, based on the rate of reciprocity in the observed network (defined as the proportion of ties that exist such that if *i* nominates *j*, then *j* also nominates *i*), as is depicted in the fourth network in Figure 4.3. A researcher could also impute the data under different assumptions, using the various imputation strategies as a way of creating bounds on the estimates (see panel (b) for options).

Especially useful for imputation are reported tie qualities. For example, when respondents report ties with an individual they see every day or with someone they consider a best friend, the alter is likely to reciprocate the relationship in surveys. Thus, that information on known ties could be used to impute missing ties.

More complicated imputation strategies are also possible. In particular, it is difficult to impute edges where nonrespondents are asymmetrically tied to someone, like F to A in Figure 4.3 (panel (a)). Such imputation problems require more model-based approaches. Here, the researcher uses as much other information as possible on individual characteristics, such as race and gender, and estimates dyadic similarities and differences to predict which missing edges are likely to exist (the models can also include network features, like reciprocity and triadic closure, which are covered in Chapter 7). This approach, however, requires knowing quite a bit about the nonrespondents, such as a past record or even all their attributes and memberships. Machine learning techniques and complex statistical models are making this sort of imputation of ties more and more feasible.

4.6.2 Posterior Tests of Robustness and Sensitivity

As a second strategy, a researcher can approach problems of measurement error and missing data by conducting *sensitivity* and *robustness tests*. For example, it is important to know if the estimates are sensitive to missing data, throwing into doubt the substantive conclusions. One simple way of gauging bias is to rely on the results of past work on missing data. Smith and Moody (2013) offer easy-to-use lookup tables for different measures and network types. Consulting these tables, researchers would find their measure of interest and identify the network(s) closest to their own network in terms of size, direction, and kind of relation. They could then determine the maximum level of missing data they could have while keeping bias below the desired level. Alternatively, regression-based models can be used to predict the level of bias (see Smith, Moody, & Morgan 2017).

Sensitivity analysis can also be used to examine measurement error. Here, one asks how much error there would need to be in the data to change the inferences made. One can even estimate likely parameters on error and then test to see if they overturn the results. Various packages can be used for this task (Frank 2000), such as *KonFound-It!* (https://jmichaelrosenberg.shinyapps.io/konfound-it/). Again, reflection on these matters assists with careful research design. Careful treatment and then honest reporting of "the best we can do" is all one can ask. From there, scholars can build and improve on one another's work.

4.7 ETHICS IN SOCIAL NETWORK DATA COLLECTION

As with other research involving human subjects, social network analysis entails careful attention to ethical and privacy concerns *prior to the data collection*. All research must be conducted in consultation with Institutional Review Boards (IRBs) to assure that subjects are protected from harm. In the case of social network analysis, harm may arise from different sources than when collecting other types of data, and having direct knowledge of a social network can, in itself, create harm. Consider how one might feel in learning that one is an isolate in a network structure, or that someone considered a friend does not reciprocate that tie. Or consider how knowledge of network structure – for example, who is popular, unpopular, or boundary-spanning in a school friendship network – could be used to try to manipulate opinion or divide students. Obviously, such network data should be collected in a manner that respects the privacy, honor, and social standing of all participants. Usually, this will mean that data should be collected in a confidentially or anonymously and kept confidential and in protected formats. At a minimum, respondents should be fully informed as to how the data will be used and what disclosure risks are implied (which can be particularly important in private consulting settings where management is using network analysis to evaluate organizational performance).

When collecting network data from individuals, as with all data based on information derived from respondents, informed consent is required. This consent would extend to parents in the case of studying children's networks. Consent may even be required in networks derived from records or online materials – that is, one must always follow the rules stated in the data use agreement, such as an API agreement for use of Twitter feeds. One complication in consent for network researchers comes from the difficulty in individuals giving consent-to-be-named. For example, when deriving a network from a snowball sampling technique, how does one go about consenting those who are named by earlier respondents? This can be particularly problematic when the data collection effort (as it often is with respondent-driven sampling

[RDS]) involves stigmatized activities, such as drug use. For roster-based data collection, such as in schools or other organizations, one can ask for consent to be placed on the roster before network data are collected. In dynamic chain-recruitment styles, researchers typically have respondents directly recruit alters. In such cases, the researcher may give out anonymous tickets for participation in RDS, or people who are named are treated as tentative contacts until they can be enrolled and informed (and data are anonymized or deleted if alter refuses to participate). Ego network designs have advantages here: they allow researchers to ask minimally identifying information (first names, initials) that are held only to aid respondent recall and are deleted after the interview.

A second problem in granting confidentiality may arise from *deductive disclosure* – that is, the capacity for informants to deduce who is who in a network based on certain known structural or node-level features. A general assumption of network boundaries is that we have identified all the relevant people in the setting, giving us a population census. This means that if somebody who obtains these data knows a person is in a sampled setting (a school or company, for example), then they also know the person is (or should be) in the data. This then could allow someone to figure out the identity of someone in the data is by the mix of attributes or pattern of network ties. For example, if one is surveying a school and knows that only one ninth-grader made the varsity basketball team, then it would be easy to find that case in the survey. Once that student is identified, a user would then have access to everything else that student reported in the survey. The best defense against deductive disclosure is masking the setting itself – never report the name of the school or company, for example. And, obviously, deductive disclosure is more likely in smaller networks; in these cases, more information will likely need to be withheld from any written report. Moreover, these issues become particularly salient in non-academic social network analysis, such as in consulting, where the data may be shared with powerful interests.

As with all data collection efforts, the precise ethical issues that may arise are highly context-dependent. When using online data, the researcher may have to sign explicit conflict of interest (COI) agreements or allow one's research partners an opportunity to embargo the publication of sensitive results. When using internal records from an organization, one may have to store data behind a firewall. When collecting audiovisual data on children at play, one may need to agree to destroy the data at some point (e.g., once the children become adults). When combining data from multiple public records, one may need to take more precautions around sensitivity as the number of sources increases and the potential for identification therefore also increases.

Network data, like much social research, is often carried out in active-use contexts for consulting or evaluation. After the 9/11 attacks, for example, the US government developed a newfound interest in the structure of terrorist networks with the goal of identifying, mapping, and "disrupting" these networks. Similarly, companies often want to know how well people work on teams and which actors provide bridges across multiple divisions, which will inform who gets promoted (or fired). As with any consulting activity, researchers will need to square the tools and technical expertise they bring to the project with their professional ethical standards, both personally and as members of professional organizations, which often have explicit codes of conduct that prohibit some sorts of activity.

We recommend that the reader consult the articles published in a 2005 special issue of the journal *Social Networks* (Volume 27, Issue 2), which was edited by Ronald Breiger. The issues we touch on here and others are covered in greater detail in those excellent articles. One should also draw on the advice and insights of one's colleagues, and especially the IRB professionals, in best practices for data collection, storage, and privacy.

4.8 SUMMARY

Thinking structurally leads to strategic data collection. No data are perfect, and all projects involve trade-offs. In this chapter, we have introduced a set of choices involved in gathering network data, using more realist and nominalist frameworks. Given the inevitable limitations of any data set, we guided readers to ways to assess the quality of their data and to engage in techniques for handling missing data. Readers should now have a pretty good sense of what it means to "think structurally" and feel ready to engage in more exploratory visualizations that frequently guide network analyses.

SUGGESTED FURTHER READING

adams, jimi. 2019. *Gathering Social Network Data*. Thousand Oaks, CA: Sage. (An outstanding recent overview of network data collection strategies.)

Bernard, H. Russell, Peter Killworth, David Kronenfeld, and Lee Sailer. 1984. "The Problem of Informant Accuracy: The Validity of Retrospective Data." *Annual Review of Anthropology* 13: 495–517. (This is one of a series of works that raised important questions on informant accuracy, leading to a focused research program on improving data collection quality.)

Birkett, Michelle, Joshua Melville, Patrick Janulis et al. 2021. "Network Canvas: Key Decisions in The Design of an Interviewer-Assisted Network Data Collection Software Suite." *Social Networks* 66: 114–24. (A significant boon to contemporary network analysis is the ability to use graphical interface devices for data collection; Network Canvas is one such tool.)

Burt, Ronald. 1984. "Network Items and the General Social Survey." *Social Networks* 6 (4): 293–339. (This paper discusses the development of the "important matters" ego network name generator and name interpreter frameworks.)

Butts, Carter. 2003. "Network Inference, Error and Informant (In)accuracy: A Bayesian Approach." *Social Networks* 25(2): 103–40. (An excellent formal treatment of the implications of error and missing data on network measures and methods.)

Domínguez, Silvia, and Betina Hollstein (eds.). 2014. *Mixed Methods Social Networks Research: Design and Applications*. New York: Cambridge University Press. (A nice combination of guide and illustration in using data sources other than traditional survey methods, including ethnographic and simulation approaches.)

Krackhardt, David. 1987. "Cognitive Social Structures." *Social Networks* 9(2): 109–34. (Cognitive social-structural data ask each member of a setting to report on the relations amongst all others, yielding a perspective on the structure of the network from each person. While time consuming, it represents a powerful way to characterize perceptions and relations simultaneously.)

Marsden, Peter V. 1990. "Network Data and Measurement." *Annual Review of Sociology* 16: 435–63. (The classic work on best practices for data collection.)

Robins, Garry, David Bright, Laurin Weissinger, and Pat Stys. 2022. "Data Collection for Social Network Research." *Social Networks* 69: 1–2. (A collection of pieces on data collection best practices.)

Tubaro, Paola, Louise Ryan, Antonio A. Casilli, and Alessio D'angelo. 2021. "Social Network Analysis: New Ethical Approaches through Collective Reflexivity. Introduction to the Special Issue of *Social Networks*." *Social Networks* 67: 1–8. (A nice collection of reflections on ethical issues in data collection.)